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RESEARCH ARTICLE

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Taking stock of consumer returns: A review and classification of the literature

Huseyn Abdulla | **Michael Ketzenberg** | **James D. Abbey**

Information and Operations Management (INFO), Mays Business School, Texas A&M University, College Station, Texas

Correspondence
James D. Abbey, Information and Operations Management (INFO), Mays Business School, Texas A&M University, MS 4217 TAMU, College Station, TX 77843.
Email: jabbey@mays.tamu.edu

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Abstract

Coincident with the rapid growth of consumer returns and their corresponding importance in the retail marketplace, academic interest in the area of consumer return policy design has significantly increased. In fact, the growth in academic publications has been tremendous, with almost half of the published works appearing within the past 6 years. The influx of new and evolving research spans across multiple disciplines and various methodologies. To provide clarity for the continued evolution of the field, we provide a comprehensive review and classification of the literature predicated on a holistic conceptual framework. The scope of the review includes all peer reviewed journal articles published prior to the end of 2018, along with any working papers cited therefrom, that specifically address (a) managerial decision-making related to return policies or (b) consumer behavior in response to such decision-making. Examining the state of the literature and practice on return policy design through the lens of a unified conceptual framework—a framework that spans both analytical and empirical research—reveals numerous managerial and theoretical opportunities for future research.

KEYWORDS

consumer returns, retail operations, return policy, systematic literature review

1 | INTRODUCTION

Consumer returns continue to grow and recently surpassed $643 billion globally (Cheng, 2015) and $369 billion in the United States alone (National Retail Federation, 2018). Academics have taken notice of the increase and significance of returns as evidenced by a recent surge in research on the topic. While research on returns extends back several decades in the economics literatures on money back guarantees and product warranties (Heal, 1977), the preponderance of the recent research largely has shifted to the disciplines of operations and marketing. Because of the rapid growth, the state of the literature regarding consumer returns remains unclear. To this end, we systematically review the literature and classify the contributions according to a holistic conceptual framework—a framework that clarifies the state of the research and also identifies theoretical gaps and opportunities for future research.

As this manuscript will highlight, research on consumer returns examines a wide range of managerial issues. The return policy establishes restocking fees, length of time allowed to make returns, channel restrictions, and more. Further, operational planning and execution activities (e.g., inventory management, product pricing, and assortment planning) influence return policy decisions or are themselves influenced by return policies. Of course, research on consumer returns also involves the management of the returned products themselves, which includes their acquisition, processing, and disposition. Return policies also influence consumer perceptions and behaviors. The consumer is intimately involved with purchase and return processes. As such, the sociological and psychometric factors that influence and

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explain consumer behavior play a significant role in setting policy and developing managerial insights. Consequently, the validity and relevance of research in this area necessitate a strong empirical foundation. Yet, as will become clear in the comprehensive review, relatively little is empirically established regarding consumer behavior in relation to return policies.

The review process began with an identification of search terms (consumer returns, return policy, money-back guarantee) with a resulting database (Google Scholar, ScienceDirect, Web of Science, JSTOR) search and abstract evaluation. To be included in the review, papers had to consider the return policy decisions of a retailer or manufacturer making direct sales. Any works not meeting the return policy decision criterion, such as those solely focused on returns from retailers to manufacturers due to buyback contracts or those that only addressed process aspects of consumer returns, were excluded. At this stage, each paper meeting the inclusion criteria also required an additional search of all works cited (i.e., citation snowballing), which led to the final set of 100 works spanning multiple decades. In sum, our review's scope includes, to the best of our knowledge, all research contributions that have been published until the end of 2018, as peer reviewed journal articles, along with any working papers cited therefrom, that specifically address (a) managerial decision-making related to return policies or (b) consumer behavior in response to such decision-making. Our inclusion criteria for the working papers are that (a) they should not be active (i.e., not undergoing peer-review but publicly available on Social Science Research Network) and (b) they should be cited in published works and therefore have influenced the subsequent research in the area. In total, the papers included in our review have been published in 43 different academic journals from various disciplines, including operations management, operations research, information systems, marketing, economics, and management.

We begin in Section 2 by introducing a conceptual framework on consumer returns. In Section 3, we transform the conceptual framework into a classification framework. We then proceed to our review and classification in Sections 4 and 5, organized separately according to the natural split that arises between analytical modeling and empirical papers. In Section 6, we conclude with a comprehensive discussion that integrates the insights from the conceptual framework, summarizes the state of the literature, characterizes the ongoing momentum of the domain, and reveals opportunities for future research.

2 | CONCEPTUAL FRAMEWORK

Investigating the anatomy of a purchase and return transaction serves as a starting point to develop a conceptual framework for classifying and reviewing the literature. Figure 1 provides a simplified view of the purchase and return process, from prepurchase to postreturn, and distinguishes epochs by the consumer's purchase and return decisions. Figure 1 is notable for providing a clear, integrated, and parallel view of the return process from both consumer and retailer perspectives, emphasizing the significant role of consumer behavior in managing consumer returns. Chircu and Mahajan (2006) conceptualize the retail transaction as a sequence of steps, including store access, search, evaluation and selection, ordering, payment, order fulfillment, and post-sales service. Each of these steps has associated costs: price-type costs (credit charges, taxes, etc.), time-type costs (waiting time, delivery time), or psychological-type costs (perceived ease of use, convenience, frustration, annoyance, anxiety, etc.).

Customers evaluate a return policy in terms of convenience for product evaluation, hassles for potential returns, learning through product experience during the return window, and the possibility of a need for continued search. Returns arise due to a multitude of factors that include uncertainties regarding product fit and valuation, along with product defects and opportunistic behavior, among others. Even prepurchase, a retailer's return policy can have a demonstrable effect on consumer behavior. More than 70% of online consumers consider return policies before making

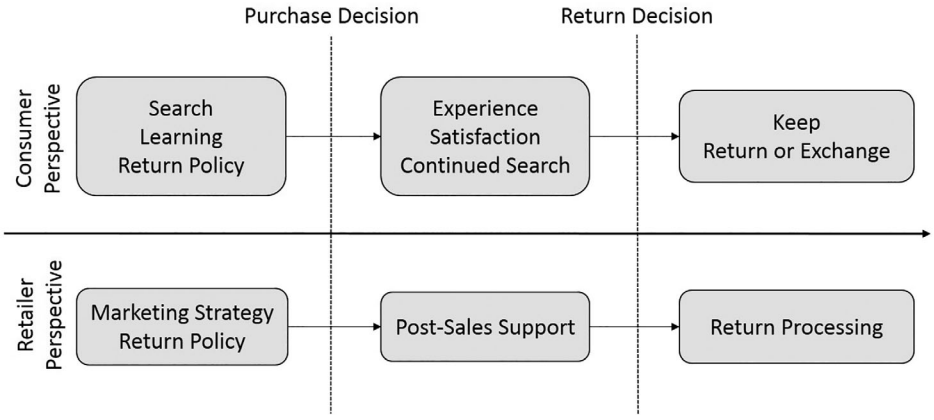


FIGURE 1 Anatomy of a purchase and return transaction

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a purchase decision (Su, 2009). Managerial decision-making with respect to the return policy, along with many other operational considerations, may affect consumer behavior at each step of the process.

From the retailer's perspective, there are distinct sets of decisions and activities to be managed prepurchase, postpurchase, and postreturn. Prepurchase, the retailer must design a return policy, provide information, set pricing, and perform all the other operational planning activities necessary for retail execution. Of course, retailer performance also continues in the context of postsales support that includes customer relationship management, warranty services, and physical management of returned products.

Collectively, from both the consumer and retailer perspectives, four broad and interrelated domains of research emerge: return policy (RP), consumer behavior (CB), planning and execution (PE), and return management (RM). These four domains are illustrated graphically in Figure 2 to form a conceptual framework. Each domain appears as a circle with interrelationships among domains denoted by their overlapping regions. We proceed by defining and characterizing the scope of research that pertains to each of the four domains, beginning with the focal RP domain.

The RP domain pertains to research that designs, describes, and prescribes return policies in the context of managerial decision-making. Return policies are generally characterized in terms of their leniency, which refers to the convenience and ease with which consumers are allowed to make returns. Some retailers have remarkably lenient return policies that allow any return for any reason, at any time, and provide a full refund of the price paid. Less lenient retailers impose restrictions, such as restocking fees, limitations on the allowable time to return, or even outright denial of returns for certain products. Research on return policy design investigates return policy leniency in various forms and analyzes the operating conditions that influence leniency choices.

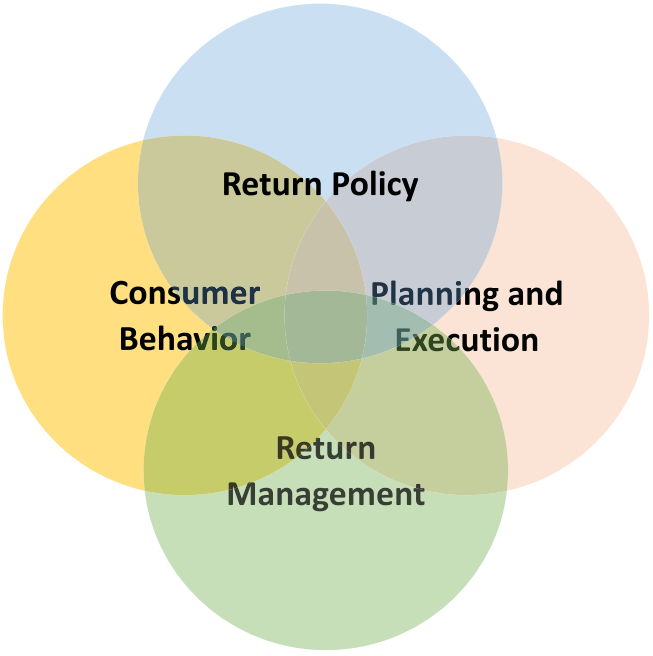


FIGURE 2 Conceptual framework [Color figure can be viewed at wileyonlinelibrary.com]

The RM domain concerns the efficient and effective acquisition, processing, and disposition of returns. RM falls under the umbrella of closed-loop supply chain management (CLSC). This considerably larger body of literature is concerned with the management of product returns at their end-of-life and end-of-use, in addition to consumer returns. For reviews of the CLSC literature, we refer the interested reader to Atasu, Guide Jr, and Van Wassenhove (2008), Souza (2013), and Govindan, Soleimani, and Kannan (2015). A retailer's return management infrastructure and practices are likely to influence return policy decisions, and vice-versa, a retailer who views its lenient return policy as an important part of the value proposition seeks ways to reduce the cost burden of returns through effective return management strategies. Thus, the inquiries in the intersection of RP and RM domains promise actionable insights for retail managers.

The CB domain explicitly considers the consumer decision-making process, with respect to both sales and returns. The primary motivation behind the study of consumer behaviors in reaction to return policies is the conjecture that sales, as well as returns, can be influenced by return policy decisions. In general, research in the intersection of CB and RP domains either (a) empirically tests relationships between return policy leniency and various constructs that reflect various cognitive, affective, and behavioral responses or (b) provides return policy prescriptions by utilizing analytical models in which consumer behaviors are operationalized through specific modeling constructs.

The PE domain captures how returns influence forward logistics and supply chain management. Our review and classification highlight several distinct areas of managerial decision-making, such as pricing, inventory, channel, assortment, competition, coordination, and quality. Moreover, decisions and actions may be taken to either reduce returns or mitigate-related costs. Whereas RM domain research addresses decision-making regarding management of returns in the reverse supply chain, PE domain research addresses decision-making regarding returns in the forward supply chain. In general, the research in the overlap of RP and PE domains studies interplay between return policy and other operational decisions in a broad variety of settings.

Collectively, the four domains of RP, RM, CB, and PE combine to form a conceptual framework that can meaningfully differentiate, and at times integrate, research contributions within the domain of consumer returns. From this perspective,

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the framework provides a foundation to classify the literature that specifically addresses consumer return policies and thereby serves as a lens to understand the state of the field.

As mentioned previously, we limit the scope of our classification solely to the focal RP domain but do include all of the intersections with the other three domains. The scope limitation is critical because incorporating the literature beyond the RP focus reveals a massive and unwieldy body of literature that would only allow a cursory examination of the literature, prohibiting a deeper understanding of the state of the research. Even with the scope restriction, this work is considerable in depth and breadth as the coming sections will demonstrate. Of course, there are many papers falling outside our scope that nonetheless make significant contributions to the study of consumer returns phenomenon. For example, there is a large body of literature on refurbishing and remanufacturing of consumer returns (Abbey & Guide Jr, 2018a). This literature fits squarely in the domain of RM and CLSC but does not involve managerial decision-making related to consumer return policies. Moreover, there are other published reviews that address this literature such as Atasu et al. (2008) and Souza (2013). In addition, there is a significant amount of published research that focuses on returns processing but does not address return policy decisions. A prime example is Griffis, Rao, Goldsby, and Niranjan (2012), which empirically investigates how a return experience and refund processing speed may influence postreturn customer behavior.

3 | TRANSFORMING CONCEPT TO CLASSIFICATION

Our objective is to review and classify the literature according to the conceptual framework. A classification will help to inform the state of the literature, clarify research contributions, and identify gaps, conflicts, and future research opportunities. This necessitates identification of various classifiers that capture key modeling choices to distinguish the contributions in the literature. In essence, the classifiers serve to transform the conceptual framework into a classification framework. A classifier plays the role of an indicator for whether a modeling construct appears within the analytical or empirical model of a given paper. Hence, the classifiers identify the modeling elements corresponding to the various domains related to CB, RP, PE, and RM. We use Su (2009) and Bonifield, Cole, and Schultz (2010)—two well-known and often cited contributions, one analytical, one empirical—as representative examples from the literature to facilitate the discussion that introduces and illustrates our classifiers. We introduce our classifiers in Table 1 for each of the four domains in our framework. Table 1 also presents the classification of Su (2009) and Bonifield et al. (2010). The presence [absence] of a check mark signifies that the respective analytical and empirical models in Su (2009) and Bonifield et al. (2010) incorporate [do not incorporate] modeling constructs corresponding to the classifier indicated by the row. We first briefly summarize Su (2009) and Bonifield et al. (2010) and then proceed to discuss the set of classifiers for each domain, beginning with the RP domain in Section 3.1.

Su (2009) examines the performance implications of return policies in a supply chain of a manufacturer and a retailer. The key contribution of this study from a return policy leniency perspective is to show that full refunds that are commonly observed in practice may be overly generous and not optimal for a retailer. Instead, the author finds that the optimal refund amount should be equal to the salvage value of a returned product. In the presence of a wholesale contract between the retailer and the manufacturer, Su (2009) shows that a partial-refund policy improves the profits of both parties relative to a full-refund policy. Next, the author studies the impact of consumer returns under different supply chain contractual mechanisms. He proves that a buy-back contract with a manufacturer may induce the retailer to adopt an overly lenient return policy. The author proposes several strategies to rectify the incentive alignment problem and achieve coordination between retailer and manufacturer.

Bonifield et al. (2010) investigate the relationship between overall leniency of a retailer's return policy and the retailer's quality through two empirical studies. In the first study, the authors content analyze the return policies of e-tailers chosen randomly from an online quality rating website and show that for nonconsumable product categories, the ratings of e-tailers are positively associated with the leniency of return policies the retailers offer. In a follow-up study, the authors show that perceived control of the online shopping experience and cognitive trust for an e-tailer moderate the relationship between return policy leniency and purchase intention.

3.1 | Return policy

There are seven classifiers for the RP domain. The first of these classifiers, money back guarantee (MBG) adoption, identifies modeling instances that investigate a retailer's binary choice between allowing returns with a full refund or disallowing them. The next five classifiers correspond to five different dimensions or levers of return policy leniency that were originally introduced as a cohesive typology of leniency by Janakiraman, Syrdal, and Freling (2016). We also require a classifier to capture the instances in which an overall degree or perception of leniency is of concern. The seven classifiers are defined in Table 1. Su (2009) investigates both MBG adoption (G) and monetary leniency (M) in the form of a refund to the customer when making

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TABLE 1 Classification framework

	Classifiers	Description	Su (2009)	Bonifield et al. (2010)
CB	Product fit and valuation uncertainty (FV)	Probability that a product fits a customer's needs and ex post realized monetary value of the purchased product to the customer.	✓	
	Customer heterogeneity (CH)	Presence of distinct segments within a customer base in terms of product fit, valuation, hassle costs, and so forth.	✓	
	Transaction costs (TC)	Customer's monetary, time, and psychological costs of returning.	✓	
	Trial period consumption (TP)	Using a product during a trial period and returning a fitting product after partial consumption based on expected utility of doing so.		
	Return timing (RT)	Time between purchasing and returning.		
	Demand and return sensitivity (DR)	Sensitivity of aggregate demand and return volume to a retailer's decisions.		
	Risk aversion (RA)	Customer's degree of risk aversion.		
	Cognitive responses (CR)	Perceptive reactions resulting from a deliberate thought process (e.g., perceived value, risk, and quality).		✓
	Affective responses (AR)	Emotional reactions to situations (e.g., regret, anger, and liking).		
	Behavioral intentions (BI)	Self-reported likelihood of engaging in certain behaviors (e.g., purchase intention, return intention, stated willingness-to-pay).		✓
	Behavioral actions (BA)	Actual outcomes of cognitive-affective responses and behavioral intentions (e.g., purchase, return, referral, manifested willingness-to-pay).		✓
RP	Money back guarantee adoption (G)	Binary decision with respect to offering an MBG.	✓	
	Monetary leniency (M)	The amount of refund or restocking fees.	✓	
	Time leniency (T)	The length of time window allowed for returns.		
	Effort leniency (F)	Degree of hassles imposed by a retailer.		
	Scope leniency (S)	Degree to which a uniform return policy applies to various products/categories.		
	Exchange leniency (X)	Whether a retailer offers cash refund or only store credit.		
	Overall leniency (O)	An overall assessment of leniency.		✓
PE	Pricing (PR)	Retailer's product price decision.	✓	
	Inventory (I)	Retailer's inventory management decisions.	✓	
	Supply chain coordination (SC)	Contractual mechanisms to coordinate the supply chain.	✓	
	Forward channel design (FD)	Retailer's optimal sales channel decisions.		
	Competition (C)	Competition between two or multiple differentiated retailers.		
	Product characteristics (PC)	Intrinsic product attributes such as quality, modularity, and so forth.		✓
	Product assortment (PA)	Retailer's product mix decisions.		
RM	Acquisition (A)	Retailer's end-of-use product collection and reverse channel design decisions.	✓	
	Processing (P)	Retailer's returns handling, restocking, recovery, and so forth.	✓	
	Disposition (D)	Retailer's returns disposition options (e.g., salvaging, returning to manufacturer, selling in a secondary market).	✓	

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a return. Bonifield et al. (2010) consider overall leniency (O) of return policies that capture leniency across multiple dimensions.

3.2 | Consumer behavior

We introduce distinct sets of classifiers to classify the analytical and empirical literature for ontological reasons. Specifically, the classifiers for the analytical literature are generated based on the constructs that are fundamental in modeling an individual customer's utility or market demand function—common operationalizations of consumer behavior in the analytical sense. This set of classifiers includes: (a) product fit and valuation uncertainty (FV), (b) customer heterogeneity (CH), (c) transaction costs (TC), (d) trial period consumption (TP), (e) return timing (RT), (f) demand and returns sensitivity (DR), and (g) risk aversion (RA). The classifiers for the empirical literature are based on four key components of consumer behavior according to the theory of buyer behavior: (a) cognitive responses (CR), (b) affective responses (AR), (c) behavioral intentions (BI), and (d) behavioral actions (BA) (Bagozzi, 1982; Howard & Sheth, 1969). We next discuss the analytical classifiers in the context of Su (2009) and follow with a discussion of the empirical classifiers in the context of Bonifield et al. (2010).

3.2.1 | Analytical classifiers

Product fit uncertainty reflects the risk that a product may fail to deliver expected quality and performance or may not fit individual customer tastes and prepurchase expectations. Valuation uncertainty relates to how much a customer values a product, typically in terms of utility, after a purchase. In the specific example of Su (2009), customers face valuation uncertainty that is only resolved after purchase. The CH classifier serves to segment the analytical literature based on assumptions regarding the ex ante (prior to purchase) characteristics of customers. Homogeneity refers to the case that all customers are identical in their intrinsic characteristics. Conversely, heterogeneity corresponds to cases in which customers have certain differentiating characteristics. In the case of Su (2009), customers are assumed to be homogeneous in the base case but heterogeneous in a model extension where product valuation uncertainty is the differentiating customer characteristic. The TC classifier corresponds to the presence of time, psychological, and monetary costs incurred by a customer making a return. These costs are often referred to as hassles in the context of the return process and include things such as filling out forms, requiring identification, and other similar costs. In the specific example of Su (2009), the author captures return hassle with a cost parameter.

Though the remaining four classifiers are not present in Su (2009), each represents an important aspect of research into RP. First, the TP classifier corresponds to product value consumed during the return window. This is an important element to modeling opportunism, which concerns the moral hazard behavior of customers who purchase items with the full ex ante intention of returning them. Ostensibly, the opportunistic customer is able to extract enough value from the product during the retailer's allowable return window such that the extractable value exceeds the costs associated with purchasing and returning the item. Second, the RT classifier corresponds to whether a model specifically addresses the length of time a customer will keep a product before making a return. Third, the DR classifier corresponds to whether, at an aggregate level, demand and return volumes are influenced by a return policy. This aggregate level approach contrasts with modeling customer utility at the individual level as in Su (2009). At the aggregate level, utility is not modeled directly. Rather, demand and return functions are generally expressed in simple linear form with respect to return policy leniency. Finally, the RA classifier relates to risk attitude and captures instances in which customers or sellers are assumed to be risk averse, as opposed to risk-neutral, which is the case in Su (2009).

3.2.2 | Empirical classifiers

The CR classifier addresses cognitive responses that include perceptions and reactions that are formed based on a conscious, deliberate thought process given available information. These may include perceptions related to risk, quality, value, effort, control, among others. For example, Bonifield et al. (2010) study "perceived control of website navigation" and "trust" to e-tailer as moderators of the relationship between return policy leniency and "perceived e-tailer quality". The AR classifier is predicated on emotional reactions to the objects, which may include feelings such as "dissonance", "liking", "satisfaction", "regret", and so on. Bonifield et al. (2010) do not consider any construct related to AR. The BI classifier relates to constructs that are based on self-reported likelihood of engaging in certain behaviors, whereas the BA classifier relates to actual, observed behavioral outcomes. In the case of Bonifield et al. (2010), the authors use customer ratings for e-tailers from an online website in the first, secondary data-based study. These customer ratings correspond to behavioral actions. In the second experimental study, the focal outcome is purchase intention, which is captured by the BI classifier.

3.3 | Planning and execution

The classifiers in the PE domain capture various operational planning and executional issues that are studied in the context of return policy decision-making. Our classification

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framework includes a set of seven classifiers. Most of these classifiers are self-explanatory. Pricing (P) concerns the determination of optimal product price. Inventory (I) addresses policies related to the management of items held in inventory and used to satisfy demand, including both the timing and quantity of replenishment, as well as storage, transportation, and handling. Supply chain coordination (SC) addresses the choice of contractual mechanisms between a retailer and a supplier or manufacturer in order to optimize supply chain performance. Forward channel design (FD) concerns the choice of one or more channels for distribution and sales, such as online and brick-and-mortar. Competition (C) addresses return policy design in the context of competing retailers or where a retailer competes with a direct-to-consumer manufacturer. Product assortment (PA) concerns the selection of products to make available for sale. Finally, product characteristics (PC) correspond to innate product characteristics such as quality, modularity, durability, and so forth. Overall, Su (2009) serves as a representative modeling instance for P, I, and SC classifiers, recognizes the C and PA classifiers as future research opportunities, but does not consider the PC classifier. Bonifield et al. (2010) show that the relationship between return policy leniency and quality ratings of e-tailers is based on whether the products that the e-tailers carry are largely consumable or nonconsumable. In essence, the moderating role of whether products are consumable or nonconsumable corresponds to an instance of PC classifier, which is the only PE-related classifier for which Bonifield et al. (2010) has an instance.

3.4 | Return management

The RM domain utilizes three classifiers that identify modeling instances of three aspects of return management: (a) acquisition (A), (b) processing (P), and (c) disposition (D). The A classifier concerns processes and decision-making with respect to collecting returns from customers. The P classifier concerns the processing of returns once acquired. Processing may be as simple as inspecting returned items for damage or may involve more intensive handling such as repackaging or refurbishing. Finally, the D classifier concerns the ultimate disposition of returned items, such as returning to the manufacturer, selling as open-box, reselling as new, and so forth. Su (2009) addresses all three RM classifiers, whereas Bonifield et al. (2010) do not address any of the RM classifiers.

To conclude, this section introduces a set of classifiers that effectively transforms the conceptual framework into a classification framework. We use Su (2009) and Bonifield et al. (2010) as representative examples to illustrate the operationalization of the framework. We now proceed to use the classifiers to organize our classification and review of the analytical (Section 4) and empirical (Section 5) literature. Two researchers conducted a binary coding of each paper based on whether there is an instance of a classifier in the paper. The initial intercoder agreement rate was 87%. The cases of disagreements were then resolved by inclusion of a third coder where the final classification status is determined through majority consensus (Durach, Kembro, & Wieland, 2017). The complete classification of both empirical and analytical literature, in an alphabetical order, appears in the Appendix A.

4 | ANALYTICAL RESEARCH

Analytical research on consumer return policies addresses the optimal return policies in a variety of operational settings and develops managerial insights. Such research commonly explores the operating conditions in which lenient or restrictive return policies are optimal. Leniency affects consumer behavior that influences both firm-level and typically supply chain-wide performance. The optimal decisions regarding a return policy often are made jointly with other operational decisions that may impact planning and execution as well as return management. Hence, all four domains of the framework are collectively integral to modeling research in this area. The classification and review of the analytical literature first examines the methods and approaches taken to model consumer behavior (Section 4.1). Next, Section 4.2 discusses findings regarding the RP classifiers and is then accompanied by the classification and review of contributions in the intersections of RP and PE (Section 4.3) as well as RP and RM domains (Section 4.4).

4.1 | Consumer behavior

This section examines the literature in the light of each CB classifier, starting with product fit and valuation uncertainty, which is then followed by a discussion of the remaining six classifiers. Though more technical in nature, the coming discussion highlights the key modeling approaches with respect to the operationalization of each classifier.

4.1.1 | Product fit and valuation uncertainty

The primary function of a return option is to reduce the risk associated with purchases. The main sources of risk are (a) product quality and performance and (b) customer tastes and prepurchase expectations. In general, the analytical literature operationalizes this risk by two parameters: fit uncertainty and valuation uncertainty. Fit uncertainty is captured by the probability that the product does not fail, has a satisfactory level of quality, or perfectly matches the customer's taste and prepurchase expectations. Valuation uncertainty is expressed in terms of a random variable whose value is realized

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postpurchase and corresponds to the monetary value (reservation price) of the purchased product to the customer. Multiple approaches are present in the literature to model fit and valuation uncertainty, largely determined by how the two parameters are related. To systematize these different approaches, we use a common set of notation motivated by the modeling approach in Yalabik, Petruzzi, and Chhajed (2005).

Let us denote the probability of product fit for a given customer with p. The existing literature refers to this parameter with a variety of names such as probability of a working product, probability of a match, likelihood of satisfaction, and so forth. We will use probability of product fit for consistency. Next, let V be a random variable with cumulative distribution function (CDF) $F(v)$, whose realization corresponds to the customer's postpurchase valuation (i.e., reservation price) of the product. The conditional CDF of the customer's valuation conditioned on the probability of fit is denoted with $F(v|p)$. Finally, let $\bar{v}$ be the upper limit of the possible realizations of V.

There are three general approaches to model product fit and valuation uncertainty. First, a stream of studies assumes $F(v|p)$ follows a Bernoulli distribution, where $V = \bar{v}$ with probability p and $V = 0$ with probability $1-p$. Hence, a customer either receives a deterministic level of valuation if the product is a match or a zero valuation otherwise. Some contributions following this approach are Davis, Gerstner, and Hagerty (1995), Moorthy and Srinivasan (1995), Fruchter and Gerstner (1999), Hsiao and Chen (2012), and the base model in Yalabik et al. (2005).

The second approach assumes that $F(v|p) = F(v)$, in which the valuation of product follows a general distribution that is independent of the fit probability p. The studies that follow this assumption usually exclude the probability of product fit as a parameter. Also, a special case of this approach is to assume $F(v|p) = v/\bar{v}$, meaning that valuation follows a general uniform distribution (i.e., $V \sim U[0, \bar{v}]$) or a standard uniform distribution in the case of $\bar{v} = 1$. Su (2009), Akçay, Boyac, and Zhang (2013), Huang, Gu, Ching, and Siu (2014), Altug and Aydinliyim (2016), and Chen and Chen (2016) use a general distribution function for valuation, whereas Davis, Hagerty, and Gerstner (1998) make the standard uniform distribution assumption to derive key insights.

The third approach separates fit uncertainty from valuation uncertainty in a similar vein to the independence approach but retains both of the parameters for explicit use in the model. These studies assume that with probability p the product is a perfect fit for the customer and that product valuation follows a uniform distribution (i.e., $V \sim U[0, \bar{v}]$), whereas with probability $1-p$ the product is not a fit and the customer's valuation is zero. This assumption allows for the fact that a product may perfectly match customer tastes and expectations; however, the customer may still have zero valuation for the product. For example, such a situation can occur if the product is no longer needed by the time the product is received. Using this approach, Hess, Chu, and Gerstner (1996), Chu, Gerstner, and Hess (1998), and Ülkü, Dailey, and Yayla-Küllü (2013) assume that product valuation follows a general uniform distribution, whereas Chen and Bell (2012), McWilliams (2012), Chen and Chen (2017a), Chen and Chen (2017b), and Yang, Chen, Chen, and Chen (2017) assume a standard uniform distribution.

A vast majority of studies assume that the return policy does not have a direct impact on a customer's postpurchase valuation of the transaction. Yet, exceptions exist, such as Ülkü et al. (2013) and Ülkü and Gürler (2018), that assume consumer valuation is distributed uniformly over $[0, \bar{v}]$, where $\bar{v}$ is equal to the sum of intrinsic valuation and positive extrinsic valuation received from the length of return time window set by the retailer. A similar approach is taken in Xu, Li, Govindan, and Xu (2015), where the incremental valuation received from return policy time leniency is added to both lower and upper bounds of the uniformly distributed intrinsic valuation parameter. Overall, this approach facilitates the assumption that customers are more likely to buy a product if the return window is longer. In Chen and Bell (2012) and Chen and Chen (2016), the authors model valuation uncertainty as the product of $V \sim U[0, 1]$ and a parameter $0 < \theta < 1$ that captures the degree that a customer accepts a nonreturnable channel (i.e., impact of a no-returns policy on valuation).

Finally, yet another approach to modeling valuation uncertainty holds that product valuation is captured as a sum of ex ante known and ex post realized components. For examples, see Liu and Xiao (2008), Shulman, Coughlan, and Savaskan (2009), Chen (2011b), and Inderst and Tirosh (2015).

4.1.2 | Customer heterogeneity

In general, analytical studies characterize the customer base by assuming that it is either homogeneous or heterogeneous. The homogeneity assumption occurs in the majority of the studies and signifies that all customers are identical in their behaviors. In contrast, literature assuming heterogeneity uses a variety of approaches in defining the nature of a heterogeneous customer base.

Moorthy and Srinivasan (1995) appear to be the first to distinguish between homogeneous and heterogeneous customer bases. In the homogeneous case, customers product valuations follow the same Bernoulli distribution as described earlier. To reflect customer heterogeneity, the authors assume that the market is segmented according to a uniform distribution $\Omega \sim U[0, 1]$, and $\Omega = \omega$ denotes the segment. Then, customer product valuations within each segment are given by ωv, where v is the reservation value of a working product. Fruchter and Gerstner

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(1999) model customer heterogeneity in terms of both product fit probability (in the base model, with a unique p_i defined for each customer) and valuation (using two customer segments with high and low product valuations).

Several recent studies use the term heterogeneous customers to highlight the assumption that customers hold uncertain valuations of the product drawn from the same probability distribution (Chen & Bell, 2012; McWilliams, 2012; Ülkü et al., 2013). Yet, many other studies do not label mere valuation uncertainty based on a single distribution as heterogeneity. To illustrate, in their base models, Su (2009) and Akçay et al. (2013) make a similar assumption as the papers above regarding customer valuation uncertainty but label such cases as homogeneous.

Overall, we find that segmenting the customer base into low and high types in terms of *ex post* product valuation is a commonly used representation of heterogeneity in the literature (Akçay et al., 2013; Hsiao & Chen, 2012; Hsiao & Chen, 2014, 2015; Su, 2009). We can refer to this representation as imperfect ex post heterogeneity as customers are still homogeneous within each segment. Additionally, there are a few studies which represent ex post customer heterogeneity in terms of a unique distribution function for product valuation of each individual customer (Anderson, Hansen, & Simester, 2009a; Chen, 2011b; Zhang, 2013). Under this assumption, we can say that customers are perfectly heterogeneous in terms of ex post valuation. In addition to heterogeneity in ex post valuation, several authors use low and high types to segment customers based on their return hassle costs (Fruchter & Gerstner, 1999; Hsiao & Chen, 2012).

Several studies assume that customers can also be *ex ante* heterogeneous on various dimensions. Fruchter and Gerstner (1999) capture ex ante heterogeneity in terms of individual probabilities of satisfaction with a purchase (i.e., product fit probability). Shulman et al. (2009) segment the customer base based on an individual taste parameter and use another parameter to reflect the likelihood that customers in a given segment know their product valuation ex ante. Shang, Ghosh, and Galbreth (2017) assume that ex ante, the customer base consists of ordinary and opportunistic customers. Finally, Xu, Li, Govindan, and Yue (2018) assume that the customer base consists of two segments. The first segment includes customers that have a higher return likelihood when they observe others who return, whereas the second segment is immune to this effect.

4.1.3 | Transaction costs

Table 2 lists the articles that explicitly include transaction costs in their models (unless otherwise stated, the papers presented in all tables follow a chronological order). The table identifies the type of transaction cost modeled and whether it is used as a parameter or decision variable. The table also reveals that considerable differences exist in the conceptual definitions of transaction costs. From a modeling standpoint, transaction costs are expressed either as a parameter or as a decision variable, with the majority of the papers falling into the former category. To the best of our knowledge, Davis et al. (1998) and Hsiao and Chen (2014) are the only two studies that model hassle cost as a decision variable, whose level is to be determined by the retailer. Several studies consider heterogeneity in hassle costs within the customer base, either based on a high- and low-type segmentation (Fruchter & Gerstner, 1999; Hsiao & Chen, 2012, 2014), or based on individual hassle cost for each customer (Anderson et al., 2009a; Chen, 2011b), whereas the majority of the studies assume homogeneous transaction costs for returning.

4.1.4 | Trial period consumption

Lenient return policies allow customers to purchase and try products during a specified time window and return the product if it does not fit or meet expectations. Trial period consumption occurs when customers extract a fraction of the product value prior to returning the product, which is key to modeling opportunism. A modeling framework addressing opportunism also needs to allow for the case that the product may be returned by a customer even when the product perfectly matches the customer's expectations. The customer may act opportunistically and return a perfectly matching product after some use if returning it for a refund provides more utility than the residual consumption value of the product beyond the trial period. Overall, this is the most common modeling paradigm to deal with opportunistic return behavior in the literature (Chu et al., 1998; Davis et al., 1995; Davis et al., 1998; Hess et al., 1996; Ülkü et al., 2013; Yalabik et al., 2005). A slightly differing approach to model opportunism is to assume that not all customers consider opportunism and there is an ex ante known segment of customers who may act opportunistically, such as in Shang, Ghosh, and Galbreth (2017). Several papers assume that such value consumption occurs only when the product is a fit (Chu et al., 1998; Davis et al., 1995; Hess et al., 1996; Ülkü et al., 2013). Although, there are a few studies that allow for some value consumption, even when there is not a fit (Hsiao & Chen, 2012; Yalabik et al., 2005).

4.1.5 | Return timing

Hess and Mayhew (1997) argue that the timing of returns is an important component to modeling consumer returns. Despite this early emphasis, however, time-related aspects in

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TABLE 2 Return policy literature involving transaction costs

Article	Term (definition, if provided)	Use	Heterogeneity
Davis et al. (1995)	Transaction cost (travel cost, lost time, and mental anxiety)	Parameter	
Moorthy and Srinivasan (1995)	Transaction cost	Parameter	
Chu et al. (1998)	Complaining cost	Parameter	
Davis et al. (1998)	Hassle (retailer imposed restrictions on returning customers)	Decision variable	
Fruchter and Gerstner (1999)	Hassle cost (traveling and confrontation costs)	Parameter	✓
Heiman, McWilliams, Zhao, and Zilberman (2002)	Return cost (inconvenience costs related to returns)	Parameter	
Yalabik et al. (2005)	Customer logistics cost	Parameter	
Matthews and Persico (2007)	Customer's return cost	Parameter	
Anderson et al. (2009a)	Customer's return cost (both monetary and psychological)	Parameter	✓
Su (2009)	Hassle cost (time, effort, and monetary resources to make a return trip to store)	Parameter	
Shulman et al. (2009)	Hassle cost (time spent in line for customer service, traveling to store)	Parameter	
Xiao, Shi, and Yang (2010)	Consumer's cost of returning	Parameter	
Chen (2011b)	Hassle cost (transportation, time value, and in-store waiting costs)	Parameter	✓
Ofek, Katona, and Sarvary (2011)	Cost of returning a mismatched product	Parameter	
Swinney (2011)	Hassle cost (travel cost of returning to a store)	Parameter	
McWilliams (2012)	Transaction cost	Parameter	
Hsiao and Chen (2012)	Hassle cost (opportunity cost of time and effort)	Parameter	✓
Ülkü et al. (2013)	Hassle cost (psychological and physical costs related to returns)	Parameter	
Zhang (2013)	Shipping, time, and effort cost	Parameter	
Hsiao and Chen (2014)	Hassle cost (transportation costs, cost of keeping an unwanted product, time values, and in-store waiting costs)	Decision variable	✓
Alptekinoğlu and Grasas (2014)	Disutility of returning (hassle cost to make a trip to store)	Parameter	
Altug and Aydinliyim (2016)	Transaction cost of returning (transportation to store, packaging and shipping, disutility from return process)	Parameter	
Chen and Chen (2017a)	Hassle cost (cost of returning a product to a store)	Parameter	✓
Yang et al. (2017)	Customer's cost of returning (transportation or shipping)	Parameter	
Chen, Chen, and Li (2018)	Customer's cost of returning	Parameter	✓
Ülkü and Gürler (2018)	Hassle cost	Parameter	

consumer return modeling are largely missing, in part due to analytical complexities that a time dimension may introduce. Hess and Mayhew (1997) propose a functional relationship between the return rate and the time-to-return, based on a hazard rate model, which is a popular method in duration modeling in marketing (Helsen & Schmittlein, 1993). Following the approach of Hess and Mayhew (1997), Jalil and Shahzad (2013) model the impact of a return policy time limit on return volumes and their variability. The authors choose a Weibull form hazard function, as the Weibull allows the authors to model returns with a truncated distribution and is flexible enough in parameters to

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facilitate different assumptions regarding return behaviors. By assuming an exponential form hazard function (i.e., constant hazard rate), Xu et al. (2015) model the expected utility of a customer from returning and not returning a product until the return deadline. Alternatively, Heiman et al. (2002) employ a learning function to capture the likelihood that a customer discovers a product misfit by the return deadline. The authors assume that customer learns the most about the product earlier in the return window and the rate of learning decreases over time. They also assume that the more time a customer has to return a product, the lower the customer's return transaction cost.

4.1.6 | Demand and return sensitivity

Several studies in the literature model aggregate market demand and return volumes as a function of a retailer's decisions, typically in a linear form. These decisions commonly include monetary leniency and price, and depending on the context, others such as product quality and degree of product modularity. The construction of aggregate market models is achieved through a set of parameters to capture the sensitivity of demand and returns to the decisions of interest. Table 3 specifies the collective papers using this assumption and details whether demand or returns (or both) arrive at an aggregate level, as well as whether the model is stochastic or deterministic. Whether return sensitivity is modeled under our classification depends on the presence of a function involving one or several retailer decisions with an associated sensitivity coefficient. For example, Mukhopadhyay and Setaputra (2004) model returns as $R = \phi + \psi r$, where ϕ is a base return amount and ψ denotes customer sensitivity to monetary leniency. Several works, such as Li et al. (2017), Heydari et al. (2017), and Batarfi et al. (2017), express returns volume only as a deterministic fraction of demand without a return-specific sensitivity parameter, and we therefore classify them as not having a return sensitivity.

In general, demand and return sensitivity across papers use consistent assumptions regarding the relationship between demand and return volumes and retailer decisions. Specifically, most works assume that demand decreases in price and increases in adoption of MBG, return policy leniency, and product modularity, whereas returns increase in return policy leniency and decrease in product quality. Among the works that are classified with a return sensitivity, none consider price sensitivity of returns, despite the earlier empirical evidence suggesting a positive association between the product price and return likelihood (Hess & Mayhew, 1997; Anderson, Hansen, Simester, & Wang, 2009b). In addition, a relationship between demand and returns, as empirically identified in Anderson et al. (2009b), is also overlooked in these works.

4.1.7 | Risk aversion

Only three instances in the literature assume that consumers are risk averse. Welling (1989) shows that if consumers have a positive degree of risk aversion, higher income consumers purchase higher quality goods, pay a higher price, and receive a higher refund if the product does not work. However, the author shows that the relationship between the optimal price-

TABLE 3 Return policy literature involving demand and return sensitivity

Article	Demand	Return	Sensitivity parameters	Stochastic
Mukhopadhyay and Setaputra (2004)	✓	✓	Price, refund amount	
Mukhopadhyay and Setaputra (2005)	✓	✓	Price, refund amount, product modularity	
Ketzenberg and Zuidwijk (2009)	✓	✓	Price, policy restrictiveness	
Xiao et al. (2010)	✓		Refund amount	✓
Liu, Choi, Yuen, and Ng (2012)	✓	✓	Price, refund amount, and product modularity	✓
Choi (2013)	✓	✓	Price, refund amount, and product modularity	✓
Choi, Liu, Ren, and Hui (2013)	✓	✓	Price, refund amount (full or none), and product modularity	✓
Li, Xu, and Li (2013)	✓	✓	Price, refund amount, and quality	
Hu, Li, and Govindan (2014)	✓		Price	✓
Yoo, Kim, and Park (2015)	✓	✓	Price, refund amount	
Altug and Aydinliyim (2016)	✓		Refund amount	✓
Li, Li, Sethi, and Guan (2017)	✓		Price, MBG adoption	
Heydari, Choi, and Radkhah (2017)	✓		Price, MBG adoption	✓
Batarfi, Jaber, and Aljazzar (2017)	✓		Price, refund amount, and migration (i.e., cannibalism)	
Difrancesco, Huchzermeier, and Schröder (2018)	✓		Return window	

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refund pair and the degree of risk aversion is ambiguous. Che (1996) demonstrates that MBG adoption is not optimal when customers are risk neutral or when retail costs are low. If the degree of customer risk aversion and high retail costs favor MBG adoption, then the retailer's profit is not affected by the degree of risk aversion. As suggested in Su (2009), Samatli-Pac, Shen, and Hu (2018) show that when customers are loss averse in addition to being risk averse, the optimal refund amount increases above the unit salvage value.

Assumptions regarding the risk attitude of consumers are typically not explicitly stated in the literature. The risk-averse consumers assumption is rare in the existing literature, appearing in only three instances. Welling (1989) shows that if consumers differ in income and have a positive degree of risk aversion, higher income consumers purchase higher quality goods, pay a higher price, and receive a higher refund should the product does not work. However, the author shows that the relationship between the optimal price-refund pair and the degree of risk aversion is ambiguous. Che (1996) demonstrates that MBG adoption is not optimal when customers are risk neutral or when retail costs are low. Whereas, customers' risk aversion and high retail costs favor mbg adoption and in this case, the retailer's profit does not get affected by the degree of risk aversion. As suggested in Su (2009), Samatli-Pac et al. (2018) show that when customers are loss averse in addition to being risk averse, the optimal refund amount increases above the unit salvage value.

4.2 | Return policy

This section classifies the research and discusses key findings with respect to return policy design. Overall, the literature shows that the degree of monetary leniency, typically expressed as a refund amount or a restocking fee, is the most extensively studied return policy decision. Only a few works investigate each of the remaining leniency levers, which leave significant opportunities for future research. The literature classified in the RP domain appears in Tables A1 and A2 within the Appendix A. The classification includes only those works that involve decisions regarding the return policy of a retailer. The decision(s) can be either endogenous, expressed as a decision variable, or exogenous, expressed as a modeling parameter. For the latter, we consider only those papers that present a sensitivity analysis with respect to the exogenously set leniency lever.

4.2.1 | Money back guarantee (MBG) adoption

In the early development of the analytical RP research, a common line of inquiry is the investigation of conditions under which it is optimal for a retailer to offer an MBG (i.e., a full-refund policy) rather than to disallow returns (i.e., a no-refund policy). For example, Davis et al. (1995) show that retailers should prefer adopting an MBG only when the customer's transaction cost of returning is low or the salvage value for returned products is high. Che (1996) finds that retailers tend to offer an MBG when customers are highly risk averse or when retail costs are high. In a more general modeling framework, Su (2009) shows that these findings are robust with respect to assumptions regarding customer valuation uncertainty. Moorthy and Srinivasan (1995) study how a high-quality retailer can adopt an MBG to signal quality and to differentiate from a low-quality retailer. More recently, McWilliams (2012) extends upon Moorthy and Srinivasan (1995) and Davis et al. (1995) to study MBG adoption as a quality signal in a duopoly. The author shows that contrary to Davis et al. (1995), a retailer can lose profit by adopting an MBG. McWilliams (2012) also shows that a low-quality retailer captures the primary benefit of quality signaling, while Moorthy and Srinivasan (1995) suggest that it is the high-quality retailer that benefits from adopting an MBG. More recently, the literature addressing the MBG decision is often coupled with other operational decisions. As such, we address such couplings within our classification at the intersection of the RP and PE domains (Section 4.3).

4.2.2 | Monetary leniency

Monetary leniency is the most commonly studied leniency lever. Modeling contexts and approaches vary significantly across studies, which makes deriving generalizable insights a challenge. In most cases, monetary leniency is defined in the context of a decision variable that represents a refund amount of the purchase price (Mukhopadhyay & Setaputra, 2004; Su, 2009; Yalabik et al., 2005). Monetary leniency can also be expressed as a restocking fee or nonrefundable charge (Hess et al., 1996; Shulman et al., 2009, 2011; Swinney, 2011). Yet another approach treats monetary leniency as a rate or percentage refund relative to the purchase price (Alptekinoğlu & Grasas, 2014; Chu et al., 1998; Mukhopadhyay & Setaputra, 2007).

A distinct feature of monetary leniency is the allowance for partial-refund policies, as opposed to the binary choice between a full refund and no refund as in the MBG classifier. Under a typical modeling framework that accounts for individual valuation uncertainty, such as in Su (2009), Shulman et al. (2009), and Akçay et al. (2013), monetary leniency affects both purchase and keep or return decisions. In effect, a customer keeps the product if her valuation is greater than the refund amount offered for a return. This implies that during a purchase decision, the customer relies on her expected utility, which is a function of the product valuation and refund amount. Under this approach, higher monetary leniency also implies a higher return probability.

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A rather consistent result reported in the literature is that full-refund policies may be overly generous and are often not optimal in a broad range of operational settings. However, the profit maximizing level of monetary leniency is open to debate and findings are context dependent. Su (2009) finds that with homogeneous customers, the optimal refund amount is equal to the salvage value, whereas heterogeneity in valuations results in more lenient refund policies. Samatli-Pac et al. (2018) demonstrate that in the case of loss-averse customers, a retailer would offer a refund that is greater than the salvage value. Xu et al. (2018) find that if the salvage value is smaller than the lower bound of customer's product valuation, the optimal refund amount should equal the lower bound, not the salvage value. Akçay et al. (2013) consider a case where a retailer sells both new and open-box items that have been returned previously. In contrast to Su (2009), they find that monetary leniency decreases if the retailer serves both high- and low-customer segments.

Chu et al. (1998) investigate partial-refund policies as a mechanism to discourage opportunism. The authors find that if customers can extract significant consumption value during the trial period and the salvage value to the retailer is low, then restocking fees are optimal. Shang, Ghosh, and Galbreth (2017) show that when opportunism is not present, the optimal refund amount equals the salvage value, as in Su (2009) and Akçay et al. (2013). However, in the presence of opportunism, the optimal refund amount is a nonincreasing function of both the extent of opportunism and the consumption benefit from opportunism.

Shulman et al. (2009) and Shulman et al. (2011) extend the inquiry of optimal monetary leniency in a competitive environment with horizontally differentiated products available for exchange. In Shulman et al. (2009), the authors reveal that, under different values of retailer attributes and customer behaviors, the optimal return policy of a monopolist may involve a full or partial refund. Shulman et al. (2011) find that, compared to a monopolist scenario, a duopoly with a product exchange option may increase the optimal restocking fee.

Overall, the majority of the articles suggest that a partial-refund policy is optimal for retailers, though this seemingly robust finding is not universal in the literature. For example, Altug and Aydinliyim (2016) discuss several conditions that may result in a full-refund equilibrium. Moreover, many mainstream retailers continue to offer full-refund policies. If in addition to individual utility benefits, lenient return policies can substantially stimulate the aggregate demand (as in demand sensitivity functions), then full refunds may be optimal. Another observation from practice is that many retailers such as Nordstrom and Macy's also offer free return shipping, which corresponds to a refund amount that is greater than the price paid for a product. Several studies, such as Fruchter and Gerstner (1999), Yalabik et al. (2005), and Matthews and Persico (2007), find such generous return policies to be optimal under certain conditions. Of course, several other studies address monetary leniency. Most of these works have unique modeling contexts tailored to investigate various PE domain trade-offs, and we discuss these works in Section 4.3 with respect to the PE classifiers.

4.2.3 | Time leniency

Our classification of the literature reveals only five studies (Difrancesco et al., 2018; Heiman et al., 2002; Jalil & Shahzad, 2013; Ülkü et al., 2013; Xu et al., 2015) in which the length of the return period is a return policy decision variable. Another study (Ülkü & Gürler, 2018) considers time leniency as a modeling parameter. Overall, the modeling contexts and trade-offs studied vary substantially across the works and often have little apparent cohesion. The unpublished working paper by Jalil and Shahzad (2013) investigates risk pooling benefits through the time leniency lever. This study shows that increasing time leniency results in reduced variability in the volume of product returns by changing customers' return behaviors.

Ülkü et al. (2013) model a retailer's price and return deadline decisions in the presence of consumer opportunism. The authors introduce an iterative procedure to jointly determine the optimal price and time leniency. Comparative statics reveal that higher time leniency leads to a higher probability of a legitimate return but also to a higher probability of an opportunistic return if the product price is below a threshold. Further, the authors show that the optimal return period may be shorter than the maximum product lifetime, if the additional utility received by customers due to time leniency is below a threshold.

Xu et al. (2015) consider a retailer deciding on a return deadline, along with refund amount, price, and order quantity. This study shows that the length of the product life cycle and consumer return rate plays crucial roles in the optimal time leniency decision. When the return rate is low, the retailer offers an indefinite return deadline. Otherwise, the optimal length of the return window depends on the product life cycle. In particular, for products with a short life cycle, the retailer chooses an indefinite return window, whereas for products with a moderately long life cycle, a fixed return deadline that is shorter than the life cycle is optimal. For products with very long life cycles, the optimal return time window is either indefinite or a fixed deadline.

Difrancesco et al. (2018) develop a model to optimize the return time window decision of a fashion retailer that has a closed-loop supply chain with refurbishing. The authors find that the optimal return window decreases as the average time until return decreases. Furthermore, the optimal return window is very large for lower values of the return rate, whereas the window sharply decreases in the return rate. The rate of

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decrease in the optimal return time window is high when products have a lower number of life cycles.

Ülkü and Gürler (2018) reveal insights regarding the relationship between a retailer's exogenously set time leniency and the retailer's order quantity, return rate, and profit, in the presence of both legitimate and opportunistic returns. First, the authors show that the retailer's order quantity and profit increase in time leniency. Second, as the return policy becomes more lenient in time, both legitimate and opportunistic return rates decrease, along with an increase in average salvage value.

4.2.4 | Effort, scope, and exchange leniency

We combine our discussion of the remaining three forms of leniency due to the dearth of research. As for effort leniency, we are aware of only three studies in the literature: Davis et al. (1998), Hsiao and Chen (2014), and Lee and Rhee (2018). Davis et al. (1998) are the first to acknowledge that most retail stores avoid partial-refund policies and restrict returns by imposing some degree of return hassle. The authors show that under certain conditions, a retailer is more likely to offer a return policy that is lenient in effort. These conditions are (a) customers cannot receive significant short term benefit from the product, (b) cross-selling margins are high, and (c) salvage values of the products returned are high.

Hsiao and Chen (2014) compare the performance of various return policy settings in the presence of product quality risk, heterogeneous valuations, and hassle costs for returning. The authors show that a return policy with a full refund and imposed hassles versus a hassle-free return policy with a restocking fee lead to the same consumer behavioral outcomes for different customer segments. However, the optimal hassle-free policy always dominates the optimal full-refund policy. Although the quality signaling view would predict that convenience in returning would be a strong signal for the seller's product quality, this study finds that optimal effort leniency decreases as the product quality increases. Finally, under certain cases, a moderate level of leniency through both monetary and effort levers is optimal. However, when customer hassle costs vary substantially across segments or when the valuation gap between the segments is large, the optimal policy does not call for the manipulation of both levers—either a full-refund or no-hassle policy will be optimal.

Lee and Rhee (2018) consider both effort and monetary leniency as decisions of a retailer who runs a resale market in addition to a conventional sales channel. The authors show that both in the presence and absence of a resale market, the optimal return policy is characterized by a hassle-free policy with a partial refund that is equal to the salvage value.

As for scope leniency, Yang et al. (2017) is the only study of which we are aware. In this study, the authors model a retailer who decides on the optimal product assortment by choosing either or both of the brands supplied by two manufacturers as well as whether to offer an MBG for each brand. The choice of return policy for the supply chain depends only on whether or not the retailer can efficiently recover the value of returned products. An MBG is optimal for both brands in the assortment provided that they have positive net salvage value.

Finally, no literature of which we are aware addresses exchange leniency. Although exchange leniency is not considered as a decision, Shulman et al. (2009), Shulman et al. (2010), and Akçay et al. (2013) model a retailer that allows customers to either return a product for a partial refund or exchange for an alternate product, which effectively corresponds to high exchange leniency.

4.3 | Return policy and planning and execution

This section examines the research on consumer returns that resides at the intersection of the RP and PE domains of our framework.

4.3.1 | Pricing

Other than return policy leniency levers, pricing is the most prevalent decision in the analytical literature on RP. This section provides a brief overview of the key findings with respect to a retailer's pricing and return policy decisions. Early papers that jointly consider return policy and pricing decisions focus on product quality signaling. For example, Shieh (1996) shows that price and MBGs together can completely reveal a monopolistic firm's private information on product quality at no additional cost. However, this study assumes that a retailer honoring an MBG does not incur any costs for handling returned products. Moorthy and Srinivasan (1995) demonstrate that an MBG can efficiently signal quality and improve profit only when a retailer's handling cost for processing returns is low. Otherwise, signaling through a higher price can be less costly.

The existing research demonstrates that a monopolistic retailer can command a price premium by offering an MBG and that this premium depends on product valuation uncertainty (Heiman et al., 2002; Matthews & Persico, 2007; Su, 2009). Fruchter and Gerstner (1999) find that in a duopoly where only one retailer offers an MBG, the retailer who offers an MBG will command a higher price and earn higher profit. McWilliams (2012) compares equilibrium prices with and without an MBG in a duopoly of quality differentiated retailers. The author finds that compared to the no-MBG case, adopting an MBG results in a price increase for both high-quality and low-quality retailers.

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The vast majority of the literature considers uniform pricing decisions for a single product. Yet, a few derive insights in a multiple product context. For example, Shulman et al. (2011) model two competing retailers, each of whom sell two horizontally differentiated products. The authors show that uncertainty in valuations of the products and differentiation between the products are key determinants of the equilibrium prices in the presence of consumer returns and exchanges. In particular, for products with little differentiation, product returns can increase prices due to limited ability to price discriminate, whereas with high differentiation, product returns reduce equilibrium prices as a result of increased restocking fees. Akçay et al. (2013) model a retailer who sells both new and open-box products that are returned and restocked. The authors show that compared to when no returns are allowed, offering an MBG without restocking returns increases new product price. Also, when the retailer resells the returned products as open-box items at a discount, the new product price increases further. However, the retailer's secondary sales opportunity lowers restocking fees.

Several works consider retailer pricing and return policy decisions for both regular and opportunistic customers. Ülkü et al. (2013) demonstrate that an increase in product price decreases the probability that a customer will keep the product but increases the probability of an opportunistic return. Further, the authors show that when price is set lower than a threshold, increasing return time leniency also increases opportunistic returns. Shang, Ghosh, and Galbreth (2017) also consider opportunism and find that the optimal price for a product decreases with the benefit from opportunism but is not increasing in the extent of opportunism.

4.3.2 | Inventory

The existing literature on consumer returns has many instances that study inventory management in the presence of returns. The typical modeling framework is similar to that of a classical newsvendor problem. Extending the problem to the consumer returns context brings in additional factors, such as return policy leniency and the salvage value of returned products, which may affect the optimal ordering strategy. Under this framework, a retailer sets price and return policy leniency and determines the order quantity in the beginning of a single-period selling horizon based on a probability distribution for uncertain demand. Once sales and returns are realized, the retailer salvages leftover new and returned products. From an inventory optimization standpoint, the objective is to obtain a closed-form solution for the optimal order quantity minimizing total cost.

We classify the RP works that address inventory management and present the resulting classification in Table 4. For the 14 inventory papers, Table 4 notes the planning horizon, describes demand characteristics, and identifies both revenue and cost streams. Because a retailer's demand model, cost, and revenue streams are the key elements to formulate and solve a newsvendor problem and modeling frameworks vary significantly in terms of these elements, we focus on these aspects in our classification. We do not include works that omit return policy decisions, such as Mostard and Teunter (2006) and Vlachos and Dekker (2003), even though these studies address inventory optimization in the context of consumer returns. Further, the classification table focuses on each paper's base model(s) and does not include modeling extensions.

Su (2009) and Ketzenberg and Zuidwijk (2009) are seminal in considering inventory optimization in the context of return policy decisions. Using a generic probability distribution to express demand uncertainty, Su (2009) finds that, compared to a full-refund return policy, a partial-refund policy induces a higher order quantity and service level, which increases a retailer's profit. As an alternative to Su (2009), Ketzenberg and Zuidwijk (2009) employ an aggregate demand model in which the demand is sensitive to price and return policy leniency. They consider a two-period selling horizon in which a retailer has an opportunity to resell returned products before salvaging at the end of the horizon. The optimal order quantity substantially decreases in price, returns policy sensitivity of demand, per unit production cost, and shows only a slight decrease in the return rate and product recovery costs.

The other studies in this area draw upon and extend the works just discussed. These studies reveal that the relationship between optimal order quantity and return policy leniency may not be always negative as suggested by Su (2009). For example, restocking returns and selling them as open-box, which is not considered by Su (2009) and Akçay et al. (2013), show that the optimal order quantity may decrease as well as increase in monetary leniency. Lee and Rhee (2018) consider several disposition options for customers: complete consumption, immediate return, and resale in an external or retailer-run secondary market after some use. The authors show that the optimal order quantity increases when a resale market is available. Ülkü and Gürler (2018) consider time leniency and opportunism and show that the optimal order quantity increases in the former and decreases in the latter. Using a time contingent return rate model, Xu et al. (2015) conclude that the optimal order quantity increases in time leniency only when the corresponding return rate is low, whereas the order quantity does not increase in case of a high return rate.

4.3.3 | Supply chain coordination

Supply chain coordination research related to consumer returns provides novel insights to the well established supply chain

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TABLE 4 Return policy literature involving inventory

Article	Event horizon	Demand	Revenue streams	Cost streams
Su (2009)	Single selling season	Random with a generic distribution	Sales of new products, salvaging unsold, and returned products	Production cost, refund for returns
Ketzenberg and Zuidwijk (2009)	Single selling season with two periods (i.e., regular sales and resale)	Linear in price and return policy leniency	Sales of new products and first-period returns, salvaging first- and second-period returns	Procurement, holding, and recovery costs, refund for returns
Xiao et al. (2010)	Single selling season	Increasing function of refund amount with a random error	Sales of new products, buyback credit for unsold products	Procurement and returns handling costs
Chen (2011a)	Single selling season	Random with a base demand plus an error term	Sales of new product, salvaging unsold, and returned products	Procurement, shortage, and returns handling costs
Akçay et al. (2013)	Single selling season	Random with a generic distribution	Sales of new and open-box products, salvaging unsold products	Procurement cost, refunds for returns
Hu et al. (2014)	Single selling season	Decreasing function of price with a random multiplier	Sales of new products, salvaging unsold, and returned products	Procurement cost, refunds for returns
Huang et al. (2014)	Single selling season	Random with a generic distribution	Sales of new products, buyback credit for unsold products	Procurement cost, refunds for returns
Xu et al. (2015)	Single selling season	Random with a generic distribution	Sales of new products, salvaging unsold, and returned products	Procurement cost, refunds for returns
Altug and Aydinliyim (2016)	Single selling season with two periods (i.e., regular sales and clearance)	Random with a generic distribution	Sales of new products, salvaging returned, and unsold products	Procurement cost, refunds for returns
Batarfi et al. (2017)	Infinite planning horizon	Deterministic with a given rate	Sales of new and refurbished products	Procurement, ordering, storage costs, and refunds for returns
Heydari et al. (2017)	Single selling season	Random with a generic distribution	Procurement cost, storage costs for returned, and unsold products	Sales of new products, buyback credit for unsold and returned products, refunds for returns
Xu et al. (2018)	Single selling season	Random with a generic distribution	New product sales, salvaging returned, and unsold products	Procurement cost, refunds for returns
Lee and Rhee (2018)	Single selling season with two periods (i.e., regular sales and resale)	Random with a generic distribution	Sales of new products, salvaging unsold or returned products	Procurement cost, refunds for returns, resale allowance paid to customers
Ülkü and Gürler (2018)	Single selling season	Random with a generic distribution	Sales of new products, salvaging legitimate and abusive returns, and unsold products	Procurement cost, loss of goodwill costs, refunds for returns

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contracting and coordination literature (Cachon, 2003). In particular, the research shows that traditional contracts may fail to achieve supply chain coordination when consumer returns are allowed by a retailer. Typically, these works involve a supply chain of a manufacturer and a retailer in which the manufacturer sets the terms of a contract. The retailer then makes decisions on price, return policy, and order quantity. All of the works consider an endogenous return policy decision, except Huang et al. (2014) that assumes an exogenous refund amount expressed as a parameter that is then examined through sensitivity analysis.

Su (2009) is the first to recognize that traditional buyback contracts with wholesale pricing may not be able to coordinate a two-echelon supply chain with consumer returns. Moreover, buyback contracts may result in inefficiently generous return policies. Two different contracts are proposed as a remedy to the coordination problem, and their effectiveness is analytically demonstrated. Samatli-Pac et al. (2018) show that the contracts proposed in Su (2009) can still coordinate the supply chain when customers are loss averse. Xiao et al. (2010) employ a demand model in which demand is sensitive to the refund amount to study a similar coordination problem and characterize the terms of a buyback contract that can coordinate the supply chain. The authors find that in the coordinated setting with a given buyback price, increasing the refund amount first increases both players' expected profits and order quantities but subsequently decreases both. Xu et al. (2015) show that the differentiated buyback contract proposed in Su (2009), which sets different buyback prices on unsold and returned units, may also fail to coordinate a supply chain if the salvage value of returned products depends on the time-to-return. To achieve coordination, the authors introduce a novel differentiated buyback contract that is contingent on the return deadline set by the retailer. Heydari et al. (2017) find that wholesale pricing and buyback contracts cannot simultaneously achieve Pareto improving coordination for both parties in a two-echelon supply chain. The authors show that a differentiated buyback contract can help achieve this goal.

Yoo et al. (2015) compare optimal prices and refunds under three contracts that may coordinate a two-echelon supply chain where the supplier has a bargaining power over the retailer. The authors find that selling prices are highest in the case of a wholesale pricing contract, followed by a buyback contract and then a quantity discount contract. Whereas, monetary leniency is greatest when a quantity discount contract is adopted, followed by a wholesale pricing contract, and then a buyback contract. Overall, the results of this study suggest that a quantity discount contract that coordinates the supply chain is the most favorable from the perspective of the supplier, as this contract leads to the lowest prices and highest return policy leniency that also stimulates sales.

4.3.4 | Forward channel design

In general, the analytical RP studies that address forward channel design investigate the conditions under which various channel structures are optimal. A typical decision model considers the choice of offering a single or dual channel (e.g., adding an online sales channel to an existing brick-and-mortar channel). In addition, channel-specific return policy leniency, as well as other operational decisions such as pricing and order quantity, is considered in these studies. No studies were found within the scope of review that considers an omnichannel environment with features such as "buy online, pick up in store" or "buy online, return to store." For each contribution, Table 5 reports study context, decision variables, and key findings from six studies under this classifier.

4.3.5 | Competition

Research addressing competition typically compares a monopoly to a duopoly. Generally, a retailer in these studies sells a single, unique product and competes with another retailer through price and return policy. Return policy decisions are usually modeled in terms of MBG adoption or the monetary leniency lever. An important assumption in modeling a competitive environment is related to the differentiation of the competing parties. Such differentiation includes product quality, market presence (i.e., established or entrant), cost advantage in returns handling or salvaging, and so forth. Both sequential (i.e., Stackelberg) and simultaneous (i.e., Nash) games are considered in these studies. Table 6 lists the relevant studies, including information regarding nature of competition, number of players, differentiating factors, and major findings.

4.3.6 | Product characteristics

Two common product characteristics studied in the analytical literature are product quality and product modularity. Retailers use return policies both as a mechanism to decrease purchase risk associated with products and as a means to signal product quality. As such, product quality choices, made by either a retailer or a manufacturer, have an impact on return policy decisions. The early literature on RP considers product quality as a source of information asymmetry between a retailer and customers and studies the quality signaling function of MBGs (Heal, 1977; Moorthy & Srinivasan, 1995; Shieh, 1996; Welling, 1989). However, this line of research assumes product quality is exogenously given and does not consider joint optimization of product quality and return policy decisions. Several studies, however, do investigate the relationship between quality decisions and monetary leniency.

Decisions regarding the degree of product modularity (i.e., customization) are commonly studied with clear trade-

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TABLE 5 Return policy literature involving forward channel design

Article	Context	Decisions	Key results
Ofek et al. (2011)	Two retailers, with the options to operate either brick-and-mortar only or through dual channels.	Price and store assistance level. Restocking fee in an extension.	There is an asymmetric equilibrium where only one firm operates an online channel yet makes lower profits than brick-and-mortar only rival. Also, there is an equilibrium where both rivals operate an online arm even if the total profits decrease. Retailers may charge restocking fees up to a maximum allowable amount.
Batarfi et al. (2017)	Single brick-and-mortar channel versus a dual-channel strategy in a closed-loop supply chain of a manufacturer, a remanufacturer, and a retailer.	Price and order quantity. Return policy is exogeneous.	When the sensitivity of demand to return policy leniency increases, demand, prices, and total profits in the supply chain increase in both single- and dual-channel systems.
Chen and Chen (2017b)	A retailer that considers adding an online channel to an existing brick-and-mortar channel.	Whether and when a dual-channel system is optimal. Whether to adopt an mbg or personalized pricing strategy (PPS) for each channel.	The retailer offers an MBG for a channel, if the net salvage value of a returned product is positive. The retailer's optimal return policy decision depends only on the efficiency in salvaging the returned products. In addition, PPS makes the online channel and MBG more attractive.
Li et al. (2017)	Dual-channel supply chain of a manufacturer and a retailer. Manufacturer can sell directly or through retailer.	Whether or not to offer full refund in each channel.	The manufacturer opts for a full-refund policy in both channels when the return rate is low. Otherwise, the manufacturer chooses a no-refund policy for both channels. For the retailer, the optimal strategy is to have a full-refund (no-refund) policy in the indirect channel and no-refund (full-refund) policy in the direct channel when the return rate is low (high).
Li et al. (2018)	A dual-channel supply chain consisting of a retailer and a manufacturer. The manufacturer sells a high-quality product indirectly through a retailer and considers opening a second, direct channel to sell relatively lower quality items.	Whether to adopt an mbg or personalized pricing strategy (pps) for each channel.	Regardless of the pricing strategy, the retailer should offer an MBG, if the net salvage value of a returned product is positive. The manufacturer benefits from having a direct channel with uniform pricing, and similar to the retailer, should offer an MBG when the net salvage value is positive.
Letizia et al. (2018)	A single manufacturer.	Whether to operate through online (i.e., direct), brick-and-mortar (i.e., through retail stores), or dual-channels. Return policy is exogeneous.	A dual-channel structure may be the optimal choice that maximizes the manufacturer's profits, due to the flexibility that the dual-channel structure provides for the manufacturer to reduce returns without charging a high restocking fee as in the case of online only channel.

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TABLE 6 Return policy literature involving competition

Article	Game type	Players and differentiating factors	Key findings
Fruchter and Gerstner (1999)	Nash	Two sellers, one offering an MBG, the other without an MBG.	The seller with an MBG charges a higher price and extracts all the expected surplus of its customers. Both sellers earn positive profits, but the seller with an MBG earns a higher profit.
Shulman et al. (2011)	Nash	Two firms, each selling two products with maximal differentiation.	Equilibrium restocking fees in a competitive environment can be higher than those charged by a monopolist. The equilibrium restocking fees increase in the extent of perceived differentiation between the products.
Ofek et al. (2011)	Nash	Two retailers, with the options to operate either brick-and-mortar only or through dual channels, choosing price and store assistance levels. Restocking fee decision is considered in an extension.	There is an asymmetric equilibrium where only one firm operates an online channel yet makes lower profits than brick-and-mortar only rival. Also, there is an equilibrium where both rivals operate an online arm even if the total profits decrease. Retailers may charge restocking fees up to a maximum allowable amount.
McWilliams (2012)	Nash	Two retailers, differentiated in quality (high vs. low) and marginal costs per sale for a product.	Duopoly profits increase with MBGs. Low-quality retailer increases profits while high-quality retailer decreases profits, compared to the case where neither retailer offers an MBG.
Chen and Grewal (2013)	Nash and Stackelberg	Two retailers, one established and offering an MBG, the other a new entrant and considering whether to offer an MBG.	When the customer acceptance of a no-refund product is low (high), the entrant retailer should offer a full-refund policy (no-refund) and set a higher (lower) price.
Inderst and Tirosh (2015)	Nash	Two firms, differentiated in quality (high vs. low).	In equilibrium, the low-quality firm offers a lower refund, even though salvage value for the products are the same and product quality is observable to customers.
Chen and Chen (2016)	Nash	Multiple retailers offering full-refund return policies but differing in other leniency levers.	As a retailer's return rate increases, it becomes less competitive as a result of increased prices and decreased market share. In equilibrium, the retailers with higher leniencies serve the customers with higher valuations, resulting in a market segmentation.
Altug and Aydinliyim (2016)	Nash	Multiple retailers with different clearence period stock availability.	If salvage value for returned products is below certain threshold, a retailer facing competition should not allow returns.

(Continues)

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TABLE 6 (Continued)

Article	Game type	Players and differentiating factors	Key findings
Yang et al. (2017)	Stackelberg	Two manufacturers, supplying a common retailer and offering two distinct brands with different production costs and fit probabilities.	If the retailer's net salvage value for a returned product supplied by either manufacturer is positive, the retailer accepts the wholesale prices set by the manufacturers and offer an MBG for both brands. This equilibrium improves profits for all parties.
Chen and Chen (2017a)	Nash	Two retailers with differing customer satisfaction rates.	Competition leads both retailers to employ an MBG and PPS. MBG adoption softens price competition and result in Pareto profit improvement, but PPS leads to intensified price competition.
Chen et al. (2018)	Nash and Stackelberg	Two retailers, differing in service quality (high vs. low) and unit costs for procurement and returns processing.	If the sum of a retailer's and customer's logistics costs is lower than the salvage value, both low and high quality retailers offer an MBG. In this case, the cost-efficient retailer will be the price leader and the competitor will be the follower, and both retailers will be better off.

offs. Sellers try to attract more customers by offering customization and lenient return policies. However, returned items are typically of low value to the seller due to their customization. Several studies investigate the trade-off between modularity and return policy leniency together with pricing decisions. Table 7 lists the studies that have instances for either of two product characteristics, identifies both the decisions that they address, and key findings. By key findings, we refer to the analytical insights regarding various trade-offs between the return policy leniency decision and other decisions, such as product quality, modularity, and price.

4.3.7 | Product assortment

Research addressing product assortment considers how the assortment decision is affected by the return policy decision and vice-versa. We are, however, aware of only four such contributions (Alptekinoğlu & Grasas, 2014; Shulman et al., 2009, 2011; Yang et al., 2017). Note that the key findings of the most recent contribution are already discussed in the context of scope leniency in Section 4.2.4.

Shulman et al. (2009) are the first to address the multiproduct case and investigates the implications with respect to monetary leniency and product pricing decisions. Customers arbitrarily pick between the two products only when the expected utility of making a purchase is positive and then make one of the three decisions: (a) return the product and leave the market, (b) exchange the purchased product with the other product, or (c) keep the purchased product. The authors show that monetary leniency in the two product case may be different from a single product case. Shulman et al. (2011) extend Shulman et al. (2009) to the case of a duopoly and show that competition leads to less generous return policies, if customers perceive strong differences between the products.

Alptekinoğlu and Grasas (2014) model a retailer who makes the assortment decision by choosing between popular, highly attractive products and eccentric, less attractive products under an exogenously given return policy. With a lenient return policy, the authors find that a retailer should decrease product variety and make the assortment mix based on popular products that have a lower probability of being returned. However, if the retailer has a restrictive return policy, the optimal assortment may consist of both popular and eccentric products.

4.4 | Return policy and return management

In general, the RM domain falls under a broader body of CLSC literature. RM research primarily investigates various trade-offs in reverse channel operations, including acquisition, processing, and disposition of consumer returns. Studies also

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TABLE 7 Return policy literature involving product characteristics

Article	Decisions	Key findings
Welling (1989)	Price, refund amount, and product quality	Both price and refund are increasing in quality. The higher the quality, the smaller the difference between price and refund amount.
Mukhopadhyay and Setaputra (2005)	Refund amount, degree of product modularity	When customer demand is more sensitive to (a) the return policy leniency, (b) the degree of product modularity, the seller offers a greater refund and higher degree of product modularity.
Mukhopadhyay and Setaputra (2007)	Product price, refund rate (as a percentage of price), product quality	Earlier in the product life cycle, quality has a steep increase and higher refund rate is provided. As price decreases over the life cycle, the return policy becomes less lenient.
Liu et al. (2012)	Product price, refund amount, and degree of product modularity	For a risk averse retailer, when baseline demand increases, optimal product price, refund rate, and degree of product modularity decreases.
Hsiao and Chen (2012)	Product price, refund amount, and degree of product risk (as an extension)	When the product quality is low, the optimal refund can exceed the product price. When the quality is relatively high, further improvement in product quality may not induce a higher product price and refund amount.
Li et al. (2013)	Product price, refund amount, product, and service quality	Pricing, refund amound, and quality decisions are mutually complementary. If the seller is a price-taker, a higher refund amount should be complemented by a higher quality.
Choi (2013)	Product price, return service charge, and degree of product modularity	Optimal return service charge increases, if degree of product modularity decreases but is independent of the sensitivity of demand to the modularity.
Choi et al. (2013)	Product price, return service charge, and degree of product modularity	When a seller offers returns, product price and modularity decisions are linearly related to each other, and price is a decreasing function of return service charge.
Yoo (2014)	Product price, refund amount, and product quality	Increasing monetary leniency cannot be optimal without enhancing product quality regardless of risk attitude. However, quality enhancement without setting a lenient return policy can be optimal.
Hsiao and Chen (2015)	Product price, refund amount, and product quality	Returns are accepted by the retailer only when the product reliability is moderate, and the valuation of the low segment is moderate. Without consumer returns, the optimal product quality always increases in the probability of a fit. With consumer returns, this relation can be reversed.

examine decisions with respect to reverse channel design in order to execute these operations, such as the allocation of returns acquisition and salvaging responsibility amongst the parties of a supply chain. Although the amount of analytical research in this domain is significant, only a few of the published studies address return policy decisions.

At a minimum, the studies at the intersection of the RP and RM domains take into account salvaging, which concerns the disposition of returns. Taking salvaging into account typically does not go beyond incorporating a single modeling parameter for the salvage value of returned products, whereas a detailed inquiry of options for and execution of salvaging is largely absent. Numerous studies show that a product's salvage value is a critical determinant of a retailer's optimal return policy choices. In particular, these studies find that retailers with more efficient salvaging options can offer more lenient policies (Altug & Aydinliyim, 2016; Davis et al., 1995; Su, 2009).

Shulman et al. (2010) focus on reverse channel design for salvaging. In particular, they investigate the optimal return policy in the context of a supply chain in which the

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TABLE 8 Theoretical basis for empirical research on return policy

Theory	Based on	Associated RP literature
Signaling	Spence (1973)	Wood (2001), d'Astous and Guèvremont (2008), Bonifield et al. (2010), Pei, Paswan, and Yan (2014), Rao, Lee, Connelly, and Iyengar (2018), Zhang, Li, Yan, and Johnston (2017), and Oghazi, Karlsson, Hellström, and Hjort (2018)
Prospect (including endowment effect)	Kahneman and Tversky (1979)	Wood (2001), Wang (2009), and Heiman, Just, McWilliams, and Zilberman (2015)
Distributive (equity) and procedural justice	Adams and Freedman (1976), Lind and Tyler (1988)	Suwelack, Hogreve, and Hoyer (2011), Bower and Maxham III (2012), and Pei et al. (2014)
Attribution	Kelley (1967)	Bower and Maxham III (2012)
Construal level	Liberman and Trope (1998)	Janakiraman and Ordóñez (2012)
Norm	Kahneman and Miller (1986)	Kim and Wansink (2012)
Cognitive dissonance	Festinger (1957)	Powers and Jack (2013)

manufacturer chooses the salvaging party: either the manufacturer itself or the retailer. The authors show that if the retailer can extract higher value from salvaging compared to the manufacturer, retailer can charge higher restocking fees than in the case in which the manufacturer is the salvaging party. Overall, this study reveals that a higher salvage value generally leads to a more lenient return policy, as also demonstrated in the earlier literature. This effect may be reversed by the manufacturer's incentive to coordinate the reverse supply chain via wholesale and buyback pricing.

There are several other return disposition issues considered in the existing literature. Two consider reselling returns. Ketzenberg and Zuidwijk (2009) investigate the case where returns are recovered at a cost and resold as new products. Alternatively, Akçay et al. (2013) consider reselling returned products as open-box (without recovery). Hsiao and Chen (2015) study the case where the retailer either salvages the returns or sends them back to the manufacturer. Nonuniform salvage values are also studied. For example, Xu et al. (2015) consider salvage value that is contingent on time-to-return and is the only study in the RP literature that considers time decay of products. Shang, Ghosh, and Galbreth (2017) and Ülkü and Gürler (2018) assume that salvage value depends on whether or not returns arise due to opportunism.

In general, returns processing in analytical models are limited to a cost parameter that captures a retailer's per unit returns processing cost. Moorthy and Srinivasan (1995) find that in absence of a positive salvage value, the return processing cost is a key determinant of the MBG adoption decision in order to efficiently signal product quality. In particular, when a retailer bears high processing costs, they should avoid offering an MBG and rely on pricing to signal quality. Swinney (2011) also considers a return processing cost and shows that the optimal restocking fee should be at least as high as this cost. Su (2009) considers both a positive salvage value and a return processing cost and demonstrates that the optimal refund amount is equal to the salvage value minus the processing cost.

On the returns acquisition side, Su (2009) shows how an arrangement where consumer returns are collected directly by a manufacturer instead of a retailer can be used to achieve supply chain coordination in a supply chain with consumer returns. Samatli-Pac et al. (2018) show that the findings in Su (2009) are robust with respect to customer loss aversion. Chen and Bell (2012) study a retailer's decision to offer a returnable, non-returnable, and dual-channel for the reverse flow of products, characterized by an MBG adoption decision. The key finding is that, for certain ranges of customer sensitivity to the prohibition to make a return, the retailer can earn a profit with a dual-channel structure that is at least equal to the profits from a single, returnable, or nonreturnable channel. The authors show that under a dual-channel structure, the retailer can segment the market by setting different prices for each channel.

5 | EMPIRICAL RESEARCH

The classification and review of the empirical literature reveals that more than half of the empirical works have been published in the last 6 years, which highlights the nascent and evolving state of the literature. Section 5.1 outlines the theoretical and methodological foundations of empirical research, which provides the basis for the discussion of key findings from the literature in Section 5.2 and Section 5.3. Section 5.2 focuses on the literature that resides at the intersection of RP and CB domains, in which the vast majority of the literature is concentrated. Section 5.3 is dedicated to the research in the intersection of return policy with PE and RM

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TABLE 9 Data collection and analysis methods in the empirical literature

Data collection/ data analysis	Analysis of variance	Regression	Structural equation modeling	Other
Survey and interview		Davis et al. (1998); Autry (2005); Heiman et al. (2015)	Mollenkopf, Rabinovich, Laseter, and Boyer (2007); Hsieh (2013); Pei et al. (2014); Oghazi et al. (2018)	Hawes and Lumpkin (1986)
Lab experiment	Van den Poel and Leunis (1999), Wood (2001), d'Astous and Guèvremont (2008), Wang (2009), Bonifield et al. (2010), Suwelack et al. (2011), Bahn and Boyd (2014), and Seo, Yoon, and Vangelova (2016)	Janakiraman and Ordóñez (2012)	Zhang et al. (2017) and Jeng (2017)	
Field experiment		Bower and Maxham III (2012), Lantz and Hjort (2013), and Rao et al. (2018)		Petersen and Kumar (2010)
Secondary data		Mixon (1999), Heim and Field (2007), Posselt, Gerstner, and Radic (2008), Anderson et al. (2009a), Bonifield et al. (2010), Zhou and Hinz (2016), Hjort and Lantz (2016), and Shang, Pekgün, Ferguson, and Galbreth (2017)		
Literature content				Heiman, McWilliams, and Zilberman (2001) and Janakiraman et al. (2016)

domains. The complete classification of the empirical literature appears in Table A3.

5.1 | Theoretical and methodological foundations

This section begins with brief descriptions of the theories used in the literature to develop hypotheses. The section then discusses data collection and analysis methods to test such hypotheses.

5.1.1 | Theoretical foundations

Empirical research in the RP domain draws upon several well-established theories from economics and psychology. Table 8 provides an easy reference for the categorization of studies based on the theoretical lenses used. To maintain an impartial approach, the categorization relies only upon the *explicitly* stated theories used in each paper's hypothesis development.

Signaling theory appears to be the most commonly employed theory. As initially proposed by Spence (1973), signaling theory addresses the reduction of information asymmetry between two parties involved in a transaction via costly signals. The theory provides a lens to study whether return policies can reduce information asymmetry between the retailer and the customer with regard to the seller quality and the product quality. Return policy leniency is a costly signal as such leniency generates significant operational costs and losses for retailers and thus serves as a valid signal in signaling theory.

Prospect theory is a behavioral economics theory that helps to explain how people choose between alternatives bearing risk when outcome probabilities are known (Kahneman & Tversky, 1979). The fundamental tenet of the theory is that people make heuristic decisions based on the

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TABLE 10 Classification of the empirical research in the intersection of RP and CB domains

RP classifiers/CB classifiers	Cognitive responses	Affective responses	Behavioral intentions	Behavioral actions
MBG adoption	Van den Poel and Leunis (1999)		Hawes and Lumpkin (1986)	Anderson et al. (2009a) and Shang, Pekgün, et al. (2017)
Monetary leniency	Wood (2001), Bower and Maxham III (2012), Pei et al. (2014), and Zhang et al. (2017)	Posselt et al. (2008) and Bower and Maxham III (2012)	Wood (2001), Wang (2009), Pei et al. (2014), Janakiraman et al. (2016), Gelbrich, Gäthke, and Hübner (2017), and Zhang et al. (2017)	Wood (2001), Heim and Field (2007), Bower and Maxham III (2012), Lantz and Hjort (2013), and Hjort and Lantz (2016)
Time leniency	d'Astous and Guèvremont (2008), Wang (2009), Suwelack et al. (2011), Janakiraman and Ordóñez (2012), Bahn and Boyd (2014), and Zhang et al. (2017)	Posselt et al. (2008) and Suwelack et al. (2011)	Wang (2009), Suwelack et al. (2011), Janakiraman and Ordóñez (2012), Heiman et al. (2015), Janakiraman et al. (2016), Rao et al. (2018), and Zhang et al. (2017)	Heim and Field (2007)
Scope leniency	Wood (2001) and Kim and Wansink (2012)		Wood (2001)	Wood (2001)
Effort leniency	Mollenkopf et al. (2007), Suwelack et al. (2011), and Janakiraman et al. (2016)	Mollenkopf et al. (2007) and Suwelack et al. (2011)	Mollenkopf et al. (2007), Suwelack et al. (2011), Janakiraman and Ordóñez (2012), and Janakiraman et al. (2016)	Heim and Field (2007)
Exchange leniency	d'Astous and Guèvremont (2008)		Heiman et al. (2015) and Janakiraman et al. (2016)	Heim and Field (2007)
Overall leniency	Bonifield et al. (2010), Hsieh (2013), Powers and Jack (2013), Jeng (2017), and Oghazi et al. (2018)	Powers and Jack (2013)	Bonifield et al. (2010), Hsieh (2013), Powers and Jack (2013), Jeng (2017), and Oghazi et al. (2018)	Petersen and Kumar (2010) and Zhou and Hinz (2016)

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TABLE 11 Empirical literature on return policy and consumer behavior—part A

Article	Leniency levers	Cognitive responses	Affective responses	Behavioral intentions	Behavioral actions	Key findings
Hawes and Lumpkin (1986)	G			Perceived purchase risk (PPR)		G $\xrightarrow{+}$ PPR
Van den Poel and Leunis (1999)	G	Purchase likelihood (PL)				G $\xrightarrow{-}$ PL
Wood (2001)	M, S	Perceived product quality (PPQ)		Continued search intention (CSI)	Return deliberation time (RDT), purchase deliberation time (PDT)	M, S $\xrightarrow{-}$ PDT M, S $\xrightarrow{+}$ PPQ M, S $\xrightarrow{\text{n.s.}}$ RDT, M $\xrightarrow{-}$ CSI
Heim and Field (2007)	M, T, F, X				Customer ratings for ease of returns and refunds (CRR)	M, F $\xrightarrow{+}$ CRR T, X $\xrightarrow{\text{n.s.}}$ CRR
Mollenkopf et al. (2007)	F	Perceived value of returns offering (PVRO)	Return satisfaction (RS)	Loyalty intentions (LI)		F $\xrightarrow{-}$ PVRO $\rightarrow +$ LI F $\xrightarrow{-}$ RS $\xrightarrow{+}$ LI
d'Astous and Guèvremont (2008)	T, X	Perceived product quality (PPQ), perceived retailer image (PRI), need for information (NI)				X, T $\xrightarrow{+}$ PRI X, T $\xrightarrow{\text{n.s.}}$ PPQ, NI
Posselt et al. (2008)	M, T		Liking (L)			M, T $\xrightarrow{+}$ L
Anderson et al. (2009a)	G				Monetary value of return option (MVRO), average category purchase rate (ACPR)	G $\xrightarrow{+}$ MVRO G $\xrightarrow{+}$ ACPR (when MVRO is high)
Wang (2009)	M, T	Endowment effect (EE), product valuation (PV)		Willingness-to-pay for product (WTPP), return intention (RI)		M, T $\xrightarrow{+}$ PV T $\xrightarrow{+}$ EE, WTPP
Bonifield et al. (2010)	O	Perceived website control (PC)		Purchase intention (PI)		O $\xrightarrow{+}$ PI (when PC is high) O $\xrightarrow{\text{n.s.}}$ PI (when PC is low)
Petersen and Kumar (2010)	O				Average purchase value (APR), average product return value (APRV), average profit per customer (APC), average number of referrals (ANR)	O $\xrightarrow{+}$ APR, APRV, APC, ANR

(Continues)

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TABLE 11 (Continued)

Article	Leniency levers	Cognitive responses	Affective responses	Behavioral intentions	Behavioral actions	Key findings
Suwelack et al. (2011)	T, F	MBG credibility (MBGC), performance risk (PR), financial risk (FR)	Anticipated regret (AR), liking (L)	Willingness-to-pay for product (WTPP), purchase intention (PI)		$T \xrightarrow{+,+,n.s.}$ MBGC, WTPP, PI $F \xrightarrow{+}$ MBGC, WTPP, PI MBGC $\xrightarrow{-,+}$ PR, $L \xrightarrow{-,+}$ WTPP MBGC $\xrightarrow{-}$ PR, FR, AR $\xrightarrow{-}$ PI MBGC $\xrightarrow{+} L \xrightarrow{+}$ PI
Bower and Maxham III (2012)	M	Perceived cost fairness (PCF), self-attribution (SA), retailer attribution (RA)	Post-return regret (PR)		Post-return spending (PS)	$M \xrightarrow{+}$ PCF (SA and RA positively moderate) $M \xrightarrow{+}$ PCF $\xrightarrow{-}$ PR $\xrightarrow{-}$ PS
Janakiraman and Ordóñez (2012)	T, F	Perceived cognitive effort (PCE), perceived physical effort (PPE)		Return propensity (RP)		$T \xrightarrow{-}$ RP (when F is low) $T \xrightarrow{n.s.}$ RP (when F is high) $T \xrightarrow{+}$ PCE, $T \xrightarrow{n.s.}$ PPE, $F \xrightarrow{-}$ PPE
Kim and Wansink (2012)	S	Upward counter-factuals (UC), downward counter-factuals (DC), product valuation (PV)				$S \xrightarrow{+}$ UC, $S \xrightarrow{+}$ DC $S \xrightarrow{+}$ PV (under pre-purchase recommendation) $S \xrightarrow{-}$ PV (under no recommendation)

RP classifiers: G: MBG adoption, M: Monetary, T: Time, F: Effort, X: Exchange, S: Scope, O: Overall leniency.

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TABLE 12 Empirical literature on return policy and consumer behavior—Part B

Article	Leniency levers	Cognitive responses	Affective responses	Behavioral intentions	Behavioral actions	Key findings
Hsieh (2013)	O	Perceived retailer opportunism (PRO), cognitive trust (CT)		Loyalty intention (LI)		$O \xrightarrow{-} PRO \xrightarrow{-} CT \xrightarrow{+} LI$
Lantz and Hjort (2013)	M				Order frequency (OF), average value of orders (AVO), average value of purchased items (AVPI), average value of returned items (AVRI), return probability (RP)	$M \xrightarrow{+} OF, RP$ $M \xrightarrow{-} AVO, AVPI$ $M \xrightarrow{n.s.} AVRI$
Powers and Jack (2013)	O	Product dissonance (PD)	Emotional dissonance (ED)	Return propensity (RP)		$O \xrightarrow{-} ED \xrightarrow{+} RP$ $O \xrightarrow{-} PD \xrightarrow{+} RP$
Bahn and Boyd (2014)	T	Assortment attractiveness (AA)				$T \xrightarrow{+} AA$ (under low variety assortment) $T \xrightarrow{-} AA$ (under high-variety assortment)
Pei et al. (2014)	M	Perceived return policy fairness (PRPF)		Purchase intention (PI)		$M \xrightarrow{+} PI, M \xrightarrow{+} PRPF$
Heiman et al. (2015)	T, X			Willingness-to-pay for return option (WTPR)		$T \xrightarrow{+} WTPR, X \xrightarrow{+} WTPR$
Hjort and Lantz (2016)	M				Average value of orders (AVO), total value of orders (TVO)	$M \xrightarrow{-} AVO, TVO$
Janakiraman et al. (2016)*	M, T, F, S, X			Purchase intention (PI), return intention (RI)		$M, F \xrightarrow{+} PI, M, F \xrightarrow{n.s.} RI$ $T, X \xrightarrow{n.s.} PI, T, X \xrightarrow{-} RI$ $S \xrightarrow{+,n.s.} RI, PI$
Zhou and Hinz (2016)	O				Sales volume (SV), number of returns (NR), overall profitability (OP)	$O \xrightarrow{+} SV, NR, OP$ (moderated by reputation)

(Continues)

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TABLE 12 (Continued)

Article	Leniency levers	Cognitive responses	Affective responses	Behavioral intentions	Behavioral actions	Key findings
Gelbrich et al. (2017)	M			Order intention (OI), keep intention (KI)		$M \xrightarrow{+} OI$ $M \xrightarrow{+} KI$ (under low shopping frequency) $M \xrightarrow{-} KI$ (under high shopping frequency)
Jeng (2017)	O	Perceived value of return policy (PVRP)		Purchase intention (PI)		$O \xrightarrow{+} PVRP \xrightarrow{+} PI$ (stronger for lesser known retailers)
Shang, Pekgün, et al. (2017)	G				Monetary value of return policy (MVRP)	$G \xrightarrow{+} MVRP$
Rao et al. (2018)	T			Willingness-to-pay for product (WTPP)		$T \xrightarrow{+} WTPP$
Zhang et al. (2017)	M, T	Perceived service quality (PSQ), perceived product quality (PPQ), perceived return difficulty (PRD)		Purchase intention (PI)		$M, T \xrightarrow{-} PRD \xrightarrow{-} PSQ \xrightarrow{+} PI$ $M \xrightarrow{+} PPQ$, $T \xrightarrow{n.s.} PPQ$
Oghazi et al. (2018)	O	Cognitive trust (CT)		Purchase intention (PI)		$O \xrightarrow{+} CT \xrightarrow{+} PI$

RP classifiers: G: MBG adoption, M: Monetary, T: Time, F: Effort, X: Exchange, S: Scope, O: Overall leniency.

* Janakiraman et al. (2016) is a metaanalytic review paper.

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potential value of losses and gains rather than on the final outcome. With its emphasis on loss aversion, this theory provides an important theoretical lens to study cognitive and behavioral responses of customers toward return policies that typically lead to monetary losses due to low monetary and/or exchange leniency (Heiman et al., 2015). Another interesting phenomenon studied under prospect theory is the endowment effect in which an individual values something already owned more than something that is not yet owned. In the context of time leniency, the endowment effect suggests that providing a longer return time window should increase a customer's valuation of a product and result in a lower return likelihood (Wood, 2001).

Two justice theories from social and organizational psychology provide a foundation to study retailer decisions relative to a customer's fairness perceptions. Distributive justice theory, also known as equity theory, posits that people value fair treatment, which motivates them to maintain relationships with organizations (Adams & Freedman, 1976). Procedural justice theory suggests that individuals are motivated not only by fairness in the distribution of outcomes, but also the processes and rules that lead to particular outcomes (Lind & Tyler, 1988). In essence, return policies are procedural statements that determine potential outcomes in case of a return event. Therefore, distributive and procedural justice theories provide a basis to study how customers perceive return policy fairness.

Attribution theory provides a framework to understand how people explain the causes of behavior and events. Although several theories of attribution exist, Kelley's Covariation Model is the most popular, which posits that people judge events caused by a particular action based on whether the action should be attributed to some characteristic of the actor or the environment (Kelley, 1967). For example, a consumer's perceptions regarding who should be held responsible for a need to return a product can moderate his or her perceptions of a retailer's return policy. Alternatively, attribution theory may also be helpful to explain customer reactions to restrictive return policy changes.

Construal level theory, developed by Liberman and Trope (1998), suggests that the temporal distance to an event alters the mental representation of events and can affect individual decisions. In particular, the theory posits that as the temporal distance gets shorter, an individual's thinking about objects and events shifts from being abstract to concrete. For example, using different scenarios of consumer choice, Liberman and Trope (1998) find that under a shortened time window, individuals rate the feasibility attributes of a service (e.g., delivery method, prices, etc.) as highly important factors, reflecting a concrete thinking. However, with a longer time window, individuals tend to focus more on desirability attributes that are more abstract (e.g., the design and color fit, user experience, etc.). In the study of return policies, a return deadline that is specified by a retailer's time leniency decision represents a temporal distance. Under high time leniency, a customer may focus more on product fit and experiential benefits, which may result in a different post-purchase outcome compared to a low leniency scenario, in which the customer would focus more on aspects such as product price and return effort. Therefore, the theory provides a basis to study how return rates are affected by the interaction of time and effort leniency, as well potential moderators in this relationship, such as product characteristics and price.

Norm theory in psychology posits that after experiencing an event, individuals develop unrealized alternative versions of the event (i.e., what the event would, could, or should have been) as a direct response (Kahneman & Miller, 1986). Kahneman and Tversky (1982) initially termed the unrealized versions of an event as counterfactuals and the mental activity of recruiting counterfactuals as counterfactual thinking. Epstude and Roese (2008) develop the functional theory of counterfactual thinking to study how the counterfactual thinking process leads to behavioral actions, based on norm theory. These theories provide a basis to develop hypotheses regarding how lenient return policies would influence responses of customers in the case of an imperfect match of a purchase that is made under lenient versus restrictive return policies.

Cognitive dissonance theory posits that inconsistencies with the personal beliefs or expectations result in cognitive dissonance. Individuals are motivated to reduce the dissonance and actively avoid situations that increase dissonance (Festinger, 1957). Cognitive dissonance theory suggests that lenient return policies can reduce the intensity of post-purchase dissonance since the purchase decisions under lenient return policies would be reversible with lower transaction costs. The study of return policies through the cognitive dissonance theoretical lens is relevant to understanding prepurchase and postpurchase, as well as postreturn behaviors of customers under high valuation uncertainty.

As seen in Table 8, less than half of the studies ground hypotheses on formal theories. Further, the existing empirical literature largely draws upon theories from behavioral economics and psychology. Therefore, significant opportunities remain for expanding the theory base, such as incorporating organizational and management theories to study the retail perspective of return policy decision-making.

5.1.2 | Methodological foundations

The empirical research on RP incorporates a variety of methods for data collection and analysis. Table 9 presents a methodological categorization. Works that involve simple descriptive statistics (Hawes & Lumpkin, 1986), meta-analysis (Janakiraman et al., 2016), and literature-driven theory development (Heiman

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et al., 2001), or works that do not explicitly specify the method used (Petersen & Kumar, 2010) appear in the "other" column.

With respect to data collection methods, laboratory experiments are most frequently used. The following observations arise from the review of experimental literature on RP: (a) all leniency levers are studied, (b) time leniency is the most commonly manipulated lever with eight studies, (c) none of the studies include manipulations of all available levers, and (d) only one study (Janakiraman & Ordóñez, 2012) explicitly studies the interaction (i.e., moderation) effect between two leniency levers (i.e., effort and time). Further, we note that most of the experimental studies follow a between-subject design with random assignment, with a few exceptions that use within subject (i.e., repeated measures) designs, such as Wood (2001) and a field experiment in Bower and Maxham III (2012).

Survey-based research is the next most commonly used technique and is largely explanatory. In most studies, the unit of analysis is at the retailer or customer level, with the latter being dominant. In fact, only Davis et al., (1998) investigates the retailer perspective with a sample of 143 retailers. The use of field experiments in RP literature is rare. Among the few contributions, Petersen and Kumar (2010) report on a field experiment that studies how several retail performance measures changed over time once an apparel and footwear retailer loosened its return policy. Lantz and Hjort (2013) conduct an online field experiment through an e-retailer's website to study customers' behavioral responses to free delivery and return shipping. As a final example, Rao et al. (2018) conduct real eBay auctions to investigate the impact of return time leniency on willingness-to-pay. As an alternative to the primary data methods outlined above, secondary data continues to gain popularity in the operations management field and research in the RP domain is no exception. The RP literature involves secondary data from a variety of sources, such as retailer transaction records (Hjort & Lantz, 2016; Shang, Pekgün, et al., 2017; Zhou & Hinz, 2016), rating websites (Bonifield et al., 2010; Heim & Field, 2007), and advertisements (Mixon, 1999).

Regarding data analysis methods, our review reveals that regression and analysis of variance techniques are most commonly used. In general, secondary data studies employ regression methods, such as ordinary least squares regression (Hjort & Lantz, 2016), logistic regression (Shang, Pekgün, et al., 2017), and multilevel modeling (Rao et al., 2018). Structural equation modeling (SEM) methods are commonly used in survey and interview research (Mollenkopf et al., 2007; Powers & Jack, 2013) but also are used to analyze experimental data (Jeng, 2017; Zhang et al., 2017). Analysis of variance techniques (e.g., ANOVA, MANOVA) are the most commonly used data analysis methods in experimental research on RP. We conjecture that the study of more complex mediation and moderation processes related to the impact of return policy leniency on various consumer behavior outcomes may increase applications of regression-based techniques to analyze experimental data. Further, we can expect that repeated measures and longitudinal experimental studies are likely to increase applications of advanced SEM techniques.

5.2 | Return policy and consumer behavior

Several empirical studies in the RP literature address how return policy leniency influences constructs related to the four empirically defined classifiers for the CB domain, that include CR, AR, BI, and BA. Table 10 categorizes the literature based on RP and CB classifiers. Predominantly, CR, BI, and BA constructs are the focus of the literature, whereas a few works consider AR constructs. In particular, our review shows that perceived product quality is the most commonly studied cognitive response, which is expected given the large number of studies that draw upon signaling theory. Liking and regret are two affective responses that are studied in two papers each. Purchase intention is the most frequently studied construct of behavioral intention. Whereas, common behavioral actions studied are average value of orders and average value of returns.

Tables 11 and 12 provide an easy reference that summarizes the key findings in the literature, in terms of relationships between different leniency levers and the constructs that operationalize the CB classifiers. The tables note the theoretical constructs that relate to each of the four empirical CB classifiers and form the basis of explicitly stated hypotheses. The last column includes the statistically significant findings related to these hypotheses. In addition, we include those statistically insignificant findings that either (a) fail to confirm earlier findings, (b) provide somewhat contradictory evidence to earlier findings, or (c) suggest that a given leniency lever may be less effective than another lever in influencing a construct. Note that two works are excluded from the tables. Heiman et al. (2001) is a conceptual piece that does not test any hypotheses that link return policy leniency to consumer behavioral constructs. Seo et al. (2016) also do not consider return policy leniency as a focal variable in their hypotheses but do involve MBG adoption as a moderator in the link between type of buying motivation (hedonic vs. utilitarian), purchase planning behavior (planned vs. unplanned), and purchase intention.

As an illustrative example for understanding Tables 11 and 12, consider Zhang et al. (2017) from Table 12. This study investigates how monetary and time leniency influence an important BI construct, purchase intention, through a mediation path of two CR constructs—perceived return difficulty and perceived service quality. From the last column, we find that higher leniency in either monetary or time leniency is associated with an increase in purchase intention. This happens through a serial mediation, in which leniency

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TABLE 13 Empirical research in RP–PE and RP–RM intersection

Article	RP							PE							RM			Key findings
	G	M	T	F	S	X	O	PR	I	SC	FD	C	PC	PA	A	P	D	
Davis et al. (1998)							✓	✓						✓			✓	There is a positive relationship between assortment variety and overall return policy leniency. There is no statistically significant relationship between retailer's salvage advantage and return policy leniency.
Mixon (1999)	✓							✓										Sellers are more likely to offer an MBG for experience goods, and the likelihood of MBG is even higher for high-price experience goods.
Van den Poel and Leunis (1999)	✓							✓										Offering both a price discount and MBG has a greater impact in terms of purchase risk relief than offering either of these individually. The authors also find that on average, MBG adoption is a more effective risk relief mechanism compared to price reduction, in terms of a positive impact on purchase likelihood.
Autry (2005)							✓									✓		There is a positive relationship between overall return policy leniency and return handling capabilities.
Posselt et al. (2008)		✓	✓					✓										A higher degree of leniency in both monetary (in the form of free return shipping) and time levers is significantly associated with higher prices.
Anderson et al. (2009a)	✓							✓					✓					The value of a return option is contingent on the product category and vary significantly across different categories.
Bonifield et al. (2010)							✓						✓					Overall return policy leniency can signal e-tailer quality, however, this effect is contingent on product characteristics. In particular, signaling effect is significant only for sellers of nonconsumable products.
Bahn and Boyd (2014)			✓											✓				Facing a high-variety assortment, time leniency is negatively associated with assortment attractiveness, whereas facing a low variety assortment, leniency is positively associated with assortment attractiveness.
Shang, Pekgün, et al. (2017)	✓												✓				✓	Salvage value (using store status and feedback volume as proxies) is a major driver behind MBG adoption decisions of eBay sellers. Competitive intensity is not a major predictor of MBG adoption decisions of eBay sellers.

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decreases perceived return difficulty and lower perceived return difficulty results in a higher perceived service quality that subsequently results in a higher purchase intention (M, T $\overset{-}{\rightarrow}$ PRD $\overset{-}{\rightarrow}$ PSQ $\overset{+}{\rightarrow}$ PI). Further, this study finds that although monetary leniency can effectively signal perceived product quality (M $\overset{+}{\rightarrow}$ PPQ), time leniency fails to do so (T $\overset{\text{n.s.}}{\rightarrow}$ PPQ). We now proceed to discuss key insights from the literature in the crossover of RP and CB domains, organized in terms of the RP classifiers, beginning with the MBG adoption classifier.

Five studies investigate the relationship between the MBG adoption and various constructs classified under CR. These studies suggest that MBG is an effective mechanism to reduce perceived purchase risk (Hawes & Lumpkin, 1986) and increases self-reported purchase likelihood (Van den Poel & Leunis, 1999). Regarding the monetary value of an MBG, Anderson et al. (2009a) find considerable variation in the value of returns across customers and categories and show that average purchase rates increase with the value of a return option. Shang, Pekgün, et al. (2017) show that offering an MBG, compared to a no-refund policy, increases the value of a return policy. However, the results suggest that the incremental monetary value of an MBG beyond the product price may not be as high as sellers may assume or as shown by Anderson et al. (2009a).

The impact of monetary leniency on consumer behavior is tested in 10 studies, using both primary and secondary data methods. For the CR classifier, several studies lend support to signaling theory by finding empirical evidence that monetary leniency signals product and service quality resulting in higher perceived value (Wang, 2009; Wood, 2001; Zhang et al., 2017). Moreover, research shows that monetary leniency positively influences perceived cost fairness (Bower & Maxham III, 2012) and policy fairness (Pei et al., 2014). For the AR classifier, research suggests that monetary leniency can stimulate liking toward a retailer's return policy (Posselt et al., 2008) and decrease postreturn regret (Bower & Maxham III, 2012). In general, there is strong empirical evidence that higher monetary leniency results in favorable customer behavioral intentions and actions from a retailer's perspective (Gelbrich et al., 2017; Lantz & Hjort, 2013; Pei et al., 2014; Zhang et al., 2017).

Ten studies explore the impact of time leniency with respect to various CB classifiers. With respect to the CR classifier, a consistent finding is that higher time leniency is associated with a higher willingness-to-pay for product and return option; however, the relationship shows a diminishing return to scale (Heiman et al., 2015; Rao et al., 2018; Suwelack et al., 2011; Wang, 2009). Research on signaling quality suggests that time leniency can signal retailer service quality but may not signal product quality (d'Astous & Guèvremont, 2008; Zhang et al., 2017). In terms of the AR classifier, research shows that time leniency can stimulate liking toward a retailer and reduce anticipated regret (Posselt et al., 2008; Suwelack et al., 2011). Regarding the BI classifier, Zhang et al. (2017) suggest that time leniency can increase purchase intention, whereas Suwelack et al. (2011) cannot find a statistically significant relationship between time leniency and purchase intention. To our knowledge, no research examines in isolation how time leniency influences behavioral actions.

Four studies address effort leniency. In terms of the CR classifier, effort leniency positively influences customer ratings for ease of returns service (Heim & Field, 2007), perceived value of returns service in general (Mollenkopf et al., 2007), and MBG credibility (Suwelack et al., 2011), but reduces performance and financial risks perceptions (Suwelack et al., 2011). Also, the studies suggest that effort leniency can also influence certain AR constructs. In particular, effort leniency can increase return satisfaction and liking but can reduce anticipated regret (Mollenkopf et al., 2007; Suwelack et al., 2011). One study investigates the interaction between effort and time leniency and finds that higher time leniency can reduce return propensity when effort leniency is also high, but may not significantly impact return propensity when effort leniency is low (Janakiraman & Ordóñez, 2012).

The relationship between scope leniency and various consumer behaviors is not studied extensively. In fact, only Wood (2001) and Kim and Wansink (2012) examine scope leniency in different ways. Wood (2001) operationalizes scope leniency in terms of whether discounted products are allowed to return. She shows that higher scope leniency increases perceived product quality and decreases purchase deliberation time. Kim and Wansink (2012) operationalize scope leniency in terms of whether products are allowed to be returned for any reason or only defective products can be returned. They show that higher scope leniency leads to a higher product valuation when a retailer offers a prepurchase recommendation but results in a lower product valuation when the retailer does not offer a recommendation.

Research that investigates exchange leniency is also limited with only three representative studies. Heim and Field (2007) show that exchange leniency is not a significant predictor of a customer's rating for a retailer's returns service convenience. d'Astous and Guèvremont (2008) conclude that exchange leniency may be effective in increasing perceived retailer image but may not signal product quality. Heiman et al. (2015) find evidence that customers value policies that offer cash refund (i.e., high exchange leniency) as opposed to store credit only (i.e., low exchange leniency).

Several studies investigate the effect of return policy leniency on consumer behavior by operationalizing leniency as an indistinguishable combination of multiple levers or as an overall leniency measure. Experimental and survey-based studies show that overall leniency of a return policy can influence various constructs associated with CR and BI classifiers. In

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particular, these studies demonstrate that lenient return policies are associated with greater trust to a retailer (Hsieh, 2013; Oghazi et al., 2018), purchase and loyalty intention (Bonifield et al., 2010; Hsieh, 2013; Jeng, 2017; Oghazi et al., 2018), and that return policy leniency can reduce dissonance and return intention (Powers & Jack, 2013). Two studies look at the relationship between overall leniency and various behavioral actions that relate to retailer performance. Zhou and Hinz (2016) find that higher leniency is associated with an increase in both sales and returns. Overall profit seems to improve, but this benefit is likely moderated by a retailer's reputation. In a field experiment, Petersen and Kumar (2010) find that increasing return policy leniency leads to an increase in average purchase and return value, average profit, and average number of referrals, all measured on a yearly per customer.

To conclude, research at the intersection of RP and CB domains provides ample empirical evidence that return policy leniency significantly impacts various cognitive and affective responses along with behavioral intentions that are antecedents of actual behavioral actions. Collectively, these findings suggest that retailers should view return policies as a key part of their overall service offering that may impact the bottom line. Finally, except for the meta-analytic review of 21 studies by Janakiraman et al. (2016), we are unaware of any study that investigates the influence of different leniency levers on consumer perceptions and/or behaviors within a unified empirical framework.

5.3 | Return policy, planning and execution, and return management

The empirical literature at the crossover of RP domain with the PE and the RM domains is rather scarce despite a few attempts in the earlier development of the literature that links return policy leniency to PE and RM concepts. Table 13 reports the few relevant studies, their classification in terms of RP, PE, and RM classifiers, and key findings.

At the RP and PE intersection, few contributions address PR, PA, PC, and C classifiers of the PE domain and MBG adoption, M, T, and O classifiers of the RP domain. As can be seen from the table, there is virtually no research that considers empirical relationships between return policy design and I, FD, and SC issues. Research in the intersection of RP and PM domains is even scarcer. To this end, we could find only three works that address the D and P classifiers of the RM domain and the G and O classifiers of the RP domain.

6 | CONCLUSIONS

This review and classification of the analytical and empirical literature on consumer return policies demonstrates that, although the count of research contributions continues to grow rapidly, the literature is still nascent and evolving, particularly within the empirical stream. Though this manuscript provides a detailed state of the literature and numerous insights of both academic and practical relevance, our conceptual framework and classification may be even more notable for signaling what remains to be addressed in future research. In particular, this section will demonstrate and stress the importance of continued triangulation in terms of theoretical, analytical, and empirical evidence as well as continued observations from practice. To this end, this section highlights essential open questions and proposes a number of possible future research directions. To streamline the discussion and provide momentum going forward, the remainder of this section examines essential topics that capture the state of the literature. Section 6.1 discusses the need for a broader conceptualization of customer heterogeneity and related implications for both research and practice. From there, we delve into theory building and the drivers of return policy design in Section 6.2, both in terms of consumer behaviors and in terms of managerial decision-making. Section 6.3 then addresses examples of how to expand insights in the domain of research on return policy design, particularly through the opportunity to extend beyond monetary leniency. Next, in Section 6.4, we address specific research opportunities that lie at the intersection of return policies with the other domains in our framework and finish in Section 6.5 by addressing needs with respect to integrating the analytical and empirical literature.

6.1 | Customer heterogeneity and implications for research and practice

The literature convincingly demonstrates that customers are heterogeneous in their return behaviors and some current and forthcoming contributions have gone so far as to classify different customer segments based on their return propensities (Foscht, Ernstreiter, Maloles III, Sinha, & Swoboda, 2013; Abbey, Ketzenberg, & Metters, 2018b). While the literature maintains a distinct focus on the differences among customers that make returns, an interesting empirical finding is the presence of a large proportion of nonreturners (Petersen & Kumar, 2010; Abbey et al., 2018b). If a customer never returns products, the literature posits that such a customer has a very high return cost, a strong preference for a retailer's products (Anderson et al., 2009b), or perhaps other psychological factors are at play (Dahl, 2016). In fact, the studies by Foscht et al. (2013) and Abbey et al. (2018b) suggest that retailers could employ differentiated return policies for different customers, penalizing heavy returners and offering incentives to infrequent returners (Gelbrich et al., 2017). In practice, we already see that some retailers are moving in this direction, as evidenced by Jet.com that offers an upfront discount to

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customers that opt out of free returns (Jet.com, 2019) and others such as Amazon who have taken action to ban customers for excessive returns (Picchi, 2018).

Excessive returners include those customers who abuse return policies in the form of opportunism or even return fraud. The difference between fraud and opportunism is not just one of the legality. At the heart of opportunism is the consumer's full intention to return a product prior to purchase and thereby obtain value at the expense of a retailer. Unlike an opportunistic return, a fraudulent return occurs when a person engages in criminal activity such as returning stolen goods—a product returned without an associated purchase transaction. Most of the academic interest has focused on opportunism. We are aware of only one active working paper (Akturk, Ketzenberg, & Yildiz, 2019) that addresses both fraud and opportunism. According to a recent survey by the National Retail Federation, opportunism, often referred to as friendly fraud, reached $5.2 billion in 2017 (National Retail Federation, 2017). In that same report, actual return fraud constitutes an even greater amount at $17.6 billion (National Retail Federation, 2017).

As stated by Petersen and Kumar (2009) and Hjort, Lantz, Ericsson, and Gattorna (2013), if different customers return different amounts, retailers need to understand the underlying relationship among customer purchase behavior, customer characteristics, and return behavior. In effect, various customer segments require differentiated managerial consideration and treatment. Fortunately, with today's technologies and loyalty cards, return policies effectively can be customized for specific customer segments and even individual customers. Best Buy currently differentiates customer classes by time leniency, offering customers that purchase higher volumes a longer period of time to make a return. In addition, several mainstream retailers contract with third parties, such as The Retail Equation, to track individual consumer return behavior in order to prevent return abuse (Safdar, 2018). Future research can examine how retailers could design differentiated policies or employ such data-driven return countermeasures to reduce abusive return behavior. Yet, how customers perceive such countermeasures or differentiated return policies is another open question. Customers may perceive such actions by retailers as promoting fairness (i.e., only penalizing abusive actors), if such countermeasures work reliably and accurately. Conversely, customers may perceive such data-driven countermeasures or customized policies as unfair, punitive, and overly invasive with regard to privacy.

6.2 | Drivers and theory of return policy design

Returns often stem from risks associated with uncertainties regarding product fit and valuation, particularly for remote purchases. Of course, a primary function of return policies is to reduce such uncertainty and perceived risk. Yet, little research examines how customers manage these risks. Indeed, the simplest means to avoid risk is to make no purchase at all. That said, if a purchase is necessary, consumers may intensify prepurchase search and learning as well as continued search and product evaluation postpurchase. Customers may also buy a basket of items, evaluate the items within the return time window, and return the items that are not satisfactory. How consumers assess return policies in relation to such risks largely remains an open empirical issue. Further, another key driver of return policy design is simply to provide a return option to enable service recovery (Mollenkopf et al., 2007). Service recovery provides the retailer with a second opportunity to satisfy customer needs and a second opportunity to generate profit (Etzel & Silverman, 1981; McCollough & Bharadwaj, 1992). Our review reveals that there is little research on service recovery options or how such service recovery relates to the return policy decision. Although not within the scope of our review, the paper by Ertekin (2018) suggests potential service recovery opportunities.

Though a fair amount of empirical work investigates consumer behavior, there is little that addresses the retailer side of the equation. Specifically, we are unaware of any research that focuses on *how* return policy decisions are made by managers in practice. In other words, what internal and external factors play a key role in return policy decision-making, either in designing new return policies or changing existing policies? Of course, the analytical research speaks to these factors, such as salvage value and product price, that influence optimal return policy design. Moreover, a number of studies links these factors to return policy leniency using secondary data, such as Davis et al. (1998) and Shang, Pekgün, et al. (2017). Yet, an empirical grounding in practice is largely absent. Further, we know little about how important the return policy decision is with respect to firm performance. How do managers align return policy decisions with their firm's overall competitive strategy? In what ways should operational and tactical decisions regarding returns and returns processes be made so that they are in alignment with the return policy and firm strategy? These represent significant opportunities for continued research.

Our classification also emphasizes the need for theory-driven empirical research on return policy design. Less than half of the already small subset of published empirical research explicitly draws upon, extends, or references established theory. Moreover, we observe that the theory base of RP research largely consists of theories from psychology and behavioral economics, which is primarily due to the dominance of studies focusing on consumer behavioral aspects of return policies. Future research, particularly in the direction of managerial decision-making, can contribute to this growth by transferring management and

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organization theories, as well as transaction cost economics theories, into the domain. We conjecture that theories of contingency, organizational information processing, organizational learning, and psychological contracts can be particularly useful in this regard. By drawing upon such theories researchers can build novel conceptual and theoretical frameworks to identify key internal factors and operating conditions that underlie return policy decisions. In turn, these frameworks can build a foundation for understanding how these factors can be linked to managerial and organizational cognition mechanisms and information processing capabilities.

6.3 | Return policy design and extending beyond monetary leniency

Analytical modeling research on return policies has a significant body of work regarding money-back guarantees and monetary leniency. Though seemingly studied at length, the review and classification reveals that significant opportunities still remain for research on monetary leniency. As an example, more research is needed to understand under what conditions a full versus partial refund is optimal. For instance, Altug and Aydinliyim (2016) attempt to demonstrate optimality of a full-refund equilibrium by revising several key modeling assumptions as discussed in Section 4.2.2. Yet, many studies conclude that partial-refund policies are optimal, as full refunds are overly generous and can subject a retailer to costly opportunistic behavior. Even without consideration of opportunism, the findings from several studies (e.g., Su (2009)) indicate partial refunds are optimal. Yet, in practice, most mainstream retailers offer full refunds for at least some extended period of time after purchase.

Continued research could incorporate empirically observed consumer behavioral reactions to return policy decisions related to full versus partial refunds when implemented in practice. One such reaction relates to unfairness perceptions and resulting outcomes due to restrictive policy changes. The importance of fairness aspects in the study of microeconomic exchanges has been long pointed out, perhaps most famously by Kahneman, Knetsch, and Thaler (1986). Empirical research, through distributive and procedural justice theoretical lenses, shows that customers perceive partial refund policies very negatively and such policies may decrease customer's positive behavioral intentions and outcomes (Bower & Maxham III, 2012; Pei et al., 2014). Further, there is significant anecdotal evidence regarding negative attitude changes toward retailers who charge restocking fees or impose restrictions to long-time lenient policies (Narvar, 2017; Price, 2018; UPS, 2017). Hence, one potential reason why major retailers maintain full-refund policies might be the potentially detrimental effect from adopting a partial refund policy in lieu of a long-established full-refund policy. In fact, many retailers seem to be aware that their overly lenient return policies hurt their bottom-line but still consider such policies an integral part of their value proposition (Dennis, 2018). In effect, retailers may choose a sub-optimal policy due to competitive or strategic considerations, though continued research needs to establish the underlying reasons.

An interesting observation from Tables A1 and A2 (see Appendix) is that out of the 67 analytical studies included in our review, only 14 involve time, effort, scope, and exchange leniencies combined. Researchers often use single-period, single product, stylized game-theoretical models that focus on optimal return policy decisions where the interplay of several operating parameters drive main results. Of course, such contributions are essential to building a foundation to the field and developing rigorous, key managerial insights. At the same time, these assumptions often are not conducive for conducting research on some of the nonmonetary return policy levers, most notably scope and exchange leniency. Consider that most modeling literature usually focuses on the single product case. By definition, scope leniency addresses the issue of return policy complexity that arises from product assortments in which characteristics across products may necessitate different return policy considerations. Relatedly, there is an apparent need for research regarding drivers of different return policies for different product categories. Regardless of the rationale, many retailers have complex return policies. For instance, the return policies for a group of 12 prominent retailers that includes Amazon, Kohl's, Best Buy, Walmart, Target, and so forth, span some 67 pages, totaling over 30,000 words (ConsumerWorld.org, 2017). Hence, return policy scope represents a significant, but largely unexplored, domain in the extant research.

The time leniency lever could also be examined in a multiple product category setting in which the objective is to develop an optimization framework to determine an optimal return policy portfolio by assigning discretized return time windows to various product categories. Shang, Ferguson, and Galbreth (2018) demonstrate category-level heterogeneity in return timing and discuss potential benefits of a category-based return policy portfolio with varying time leniency. Thus, significant opportunities remain for research on nonmonetary policy levers. Research that addresses more than a single return policy lever is rare, providing opportunities to explore potential interactions amongst the different leniency levers as well as establishing their relative importance in setting policy.

More generally, research is needed on the analysis of product assortments and return policy design given the many varied consumer behavioral reactions. For example, consumers typically buy baskets of goods and may have different return propensities for different items in a basket. How does the return policy affect basket size and product selection? For instance, retailers often offer free shipping for a minimum purchase size. Consumers may fill the cart with items in order

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to meet the threshold and then later return those "filler" items. Cachon, Gallino, and Xu (2018) is an active working paper on this topic that highlights the need to also address return policy design issues. As another example, consider that customers often engage in sampling behavior in which they will buy multiple products with the express intent of trying them all and returning all but the one or two that fit best—a practice encouraged and exemplified by Amazon's Prime Wardrobe service. Though these types of behaviors are common and prevalent, the current modeling paradigms are unable to capture these effects when examining a single product.

Overall, a fundamental premise underlying virtually all models addressing return policy decisions is that returns are costly (e.g., requiring that the salvage value of a return is less than the procurement cost). Of course, returns can indeed be quite costly as evidenced by a plethora of industry sources (Boyajian, 2018; Dennis, 2018; Enright, 2013; National Retail Federation, 2017). Nevertheless, growing evidence shows that returning customers are more profitable than nonreturning customers (Hjort et al., 2013; Abbey et al., 2018b). In fact, customer profitability can actually increase as returns propensity increases, at least up to a threshold (Petersen & Kumar, 2010; Abbey et al., 2018b). Apparently, customers who value return policy leniency will purchase more and exhibit greater loyalty than their nonreturning counterparts. Thus, there is an opportunity for analytical research focused at the customer level that can provide important insights not available at the transaction level. As noted by Mollenkopf, Frankel, and Russo (2011), Petersen and Kumar (2010), and Abbey et al. (2018b), managing costs as well as customer relationships are both essential to effectively managing returns. Extending the modeling paradigm to include multiple periods or otherwise allowing multiple purchases and multiple returns is a key means to enable analysis at the customer level and for managing customer relationships over a lifetime of transactions. Moreover, such consumers appear to continually repeat such return transactions—behavior that has inspired the label *serial returner* (Samorani, Alptekinoglu, & Messinger, 2016). Yet, the single period models commonly used in the analytical modeling literature are not conducive to studying such characteristics. As such, the modeling of serial return behavior represents a motivating example to extend the modeling paradigm to include multiperiod models. Similarly, time leniency, time-to-return, as well as the decay in the value of product returns over time, also serve as additional catalysts for the study of multiperiod models on consumer returns. Such expanded analyzes may provide new insights into the optimality of partial or full refunds.

6.4 | Research at the intersection of domains

There is a paucity of research on return policy design that intersects the RM and PE domains. Consider for example, research at the intersection of the RM domain. The empirical results in the literature indicate that post-return handling and management is an important element to return policy design (Bower & Maxham III, 2012; Petersen & Kumar, 2010). In the modeling literature, however, return management is typically abstracted to simply assuming a fixed salvage value for returns. Shulman et al. (2010), Akçay et al. (2013), and Hsiao and Chen (2015) are notable exceptions that delve into decision-making aspects of return processing or disposition.

The cost and timeliness of acquiring and processing returns should have implications with respect to return policy design. For example, research has shown that the value of product returns can decay over time. Hence, one would expect time leniency to be affected by the rate of the decay, as modeled in Xu et al. (2015), but there may also be implications with respect to the other leniency levers. Hence, there remains an opportunity to empirically investigate the relative importance and influence that these levers have on consumer behavior, particularly with regard to return timing. These latter observations also indicate additional research opportunities arising at the intersection of the CB and RP domains.

Research on disposition options for returns in the context of return policy decision-making is limited. Many returns are never opened, remain effectively as good as new, and can simply be put back on the shelf. Yet, other returns can only be liquidated. In fact, a variety of disposition options exist, including: resell as new, warranty replacement, resell in secondary markets, resell as open-box item, salvage, dispose, and liquidate to third parties. Indeed, many retailers employ a variety of methods to dispose of their returns. Joint examination of a firm's disposition capabilities and return policy decisions, perhaps also incorporating planning and execution insights, offers yet another research opportunity (Autry, 2005). For example, when returns can be resold as new or open-box, the required quantity of new products is reduced (Akçay et al., 2013; Ketzenberg & Zuidwijk, 2009). When returns can only be salvaged, the required quantity of new products increases (Su, 2009). This simple comparison illustrates different outcomes for planning and execution (inventory optimization) with respect to differing disposition options and highlights the need for research into how a return policy interacts with these operational elements.

As with RM, there remains a need for research at the intersection of the RP and PE domains. Indeed, there is a growing body of literature that considers numerous PE-related issues in the context of product returns, but most of this literature does not address decision-making regarding the return policy itself. Consider for example that there are a number of operational tools, relevant to planning and execution activities, whose effectiveness in reducing purchase risks (i.e., increasing sales) and mitigating returns

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are empirically documented. For example, Gallino and Moreno (2018) demonstrate in a randomized field experiment that virtual fitting-room technologies can increase sales and decrease likelihood of returns, mostly due to decreasing the number of multiple-ordering customers. De, Hu, and Rahman (2013) investigate the effect of different product-oriented web technology features (i.e., zooming, alternative product photos, color swatch) on returns and conclude that not all types of product information provision techniques are likely to reduce returns. In practice, the implementation of such technologies requires significant investments from retailers. Hence, future research could assess the conditions under which firms can generate value from such technology investments and how such investments enable changes in return policies.

6.5 | Integrating empirical and analytical research

A final point of our study relates to opportunities for methodological triangulation and integrating analytical and empirical research (Singhal & Singhal, 2012). Integration is not merely about citing earlier analytical (empirical) findings in a new empirical (analytical) study. Integration also involves (a) conducting empirical-analytical hybrid studies, such as Davis et al. (1998), (b) empirical testing of insights from analytical literature, and (c) incorporating new findings from empirical literature in developing new analytical models. We elaborate on points (b) and (c) below.

Our classification is noteworthy by identifying several new and significant findings from the empirical literature on CB that have not yet been integrated into analytical studies. For instance, though a large proportion of customers apparently do not make a return (Petersen & Kumar, 2009; Abbey et al., 2018b), many customers value the return policy itself and are therefore generally willing to pay more for return policy leniency (Heiman et al., 2015; Rao et al., 2018; Wang, 2009). Further, an item will tend to exhibit a lower return rate when offered at a lower price than at a higher price (i.e., a within-item price effect) (Shang et al., 2018). Similarly, items in an assortment offered at higher prices will tend to also exhibit higher return rates than other items in the same assortment offered at lower prices (i.e., a between item price effect). We also note that the probability of a return decreases in the time since purchase (i.e., the endowment effect). Finally, as mentioned previously, customers who return more are *generally* more profitable than customers who return less. Addressing these empirical findings in the context of analytical models holds much promise for enhancing our knowledge concerning optimal return policy decision-making.

Just as empirical research is capable of informing analytical modeling research, insights generated from the modeling literature can also benefit from research triangulation from the empirical literature. Consider the relationship between competition and return policy leniency. Shulman et al. (2011), Ofek et al. (2011), and Altug and Aydinliyim (2016) suggest that competition may result in lower policy leniency, whereas Chen and Chen (2017a) and Chen et al. (2018) suggest otherwise. Alternatively, Shang, Pekgün, et al. (2017), using eBay data, do not find a relationship between competitive intensity and MBG adoption. Hence, empirical investigation through behavioral experiments or other methods may help to clarify how retailers integrate competitive issues into their return policy decisions and help to identify the significant moderating factors.

As this concluding section makes clear, though the extensive classification and review revealed a significant body of research, numerous triangulation opportunities across the analytical and empirical domains remain. The above represents a mere sampling of the myriad of future directions available to researchers in the RP domain. Nonetheless, we hope that this discussion will be thought provoking and provide a first step toward understanding and seizing upon these future research opportunities.

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ORCID

Huseyn Abdulla https://orcid.org/0000-0002-8035-7119
Michael Ketzenberg https://orcid.org/0000-0003-4890-8328
James D. Abbey https://orcid.org/0000-0003-3722-4692

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APPENDIX A

TABLE A1 Classification of the analytical literature—Part A

Article	CB							RP							PE							RM		
	PV	CH	TC	TP	RT	DR	RA	G	M	T	F	S	X	O	PR	I	SC	FD	C	PC	PA	A	P	D
Akçay et al. (2013)	✓	✓							✓				✓		✓	✓								✓
Alptekinoğlu and Grasas (2014)	✓		✓					✓	✓												✓			✓
Altug and Aydinliyim (2016)	✓		✓						✓						✓	✓			✓					✓
Anderson et al. (2009a)	✓	✓	✓					✓							✓					✓				
Batarfi et al. (2017)						✓			✓						✓	✓		✓					✓	✓
Che (1996)	✓						✓	✓							✓									
Chen and Chen (2017a)	✓		✓					✓							✓				✓				✓	✓
Chen and Chen (2017b)	✓		✓					✓							✓			✓					✓	✓
Chen and Bell (2012)	✓							✓							✓	✓						✓		✓
Chen and Chen (2016)	✓													✓	✓				✓				✓	✓
Chen and Grewal (2013)	✓							✓							✓				✓					✓
Chen et al. (2018)	✓		✓					✓							✓				✓				✓	✓
Chen (2011b)	✓	✓	✓						✓						✓									✓
Choi (2013)						✓			✓						✓					✓				✓
Choi et al. (2013)						✓		✓							✓					✓				
Chu et al. (1998)	✓		✓	✓					✓						✓									✓
Davis et al. (1995)	✓		✓	✓				✓							✓									✓
Davis et al. (1998)	✓		✓								✓				✓									✓
Difrancesco et al. (2018)					✓	✓			✓		✓											✓	✓	✓
Fruchter and Gerstner (1999)	✓	✓	✓					✓	✓						✓				✓					✓
Heal (1977)									✓						✓									
Heiman et al. (2002)			✓		✓				✓	✓					✓									
Hess et al. (1996)	✓		✓	✓					✓						✓									✓
Heydari et al. (2017)						✓		✓							✓	✓	✓						✓	✓
Hsiao and Chen (2012)	✓	✓	✓					✓	✓						✓					✓				✓
Hsiao and Chen (2014)	✓	✓	✓	✓				✓			✓				✓									✓

(Continues)

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TABLE A1 (Continued)

Article	CB							RP							PE							RM		
	PV	CH	TC	TP	RT	DR	RA	G	M	T	F	S	X	O	PR	I	SC	FD	C	PC	PA	A	P	D
Hsiao and Chen (2015)	✓	✓		✓					✓						✓					✓		✓		✓
Hu et al. (2014)	✓					✓			✓						✓	✓	✓							✓
Huang et al. (2014)	✓								✓						✓	✓	✓							✓
Inderst and Tirosh (2015)	✓								✓						✓				✓					✓
Jalil and Shahzad (2013)					✓	✓				✓														
Ketzenberg and Zuidwijk (2009)	✓					✓								✓	✓	✓							✓	✓
Lee and Rhee (2018)	✓			✓					✓		✓				✓	✓						✓		✓

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TABLE A2 Classification of the analytical literature—Part B

Article	CB							RP							PE							RM		
	PV	CH	TC	TP	RT	DR	RA	G	M	T	F	S	X	O	PR	I	SC	FD	C	PC	PA	A	P	D
Letizia, Pourakbar, and Harrison (2018)	✓								✓						✓			✓						✓
Li et al. (2013)						✓			✓						✓					✓				
Li et al. (2017)						✓		✓							✓			✓						✓
Li, Chen, Liang, and Chen (2018)	✓		✓					✓							✓			✓					✓	✓
Liu and Xiao (2008)	✓	✓							✓						✓	✓								
Liu et al. (2012)						✓			✓						✓					✓				
Liu, Mantin, and Wang (2014)	✓					✓			✓							✓	✓						✓	✓
Mann and Wissink (1988)	✓								✓						✓									✓
Matthews and Persico (2007)	✓		✓						✓						✓									✓
McWilliams (2012)	✓		✓					✓							✓				✓				✓	✓
Moorthy and Srinivasan (1995)	✓	✓	✓					✓							✓								✓	
Mukhopadhyay and Setaputra (2004)						✓			✓						✓									
Mukhopadhyay and Setaputra (2005)						✓			✓						✓					✓				
Mukhopadhyay and Setaputra (2007)						✓			✓						✓					✓				
Ofek et al. (2011)																								
Samatli-Pac et al. (2018)	✓						✓		✓						✓	✓	✓					✓	✓	✓
Shang, Ghosh, and Galbreth (2017)	✓	✓		✓					✓						✓									✓
Shieh (1996)	✓							✓	✓						✓									
Shulman et al. (2009)	✓	✓	✓						✓				✓		✓									✓
Shulman, Coughlan, and Savaskan (2010)	✓	✓							✓				✓		✓							✓		✓
Shulman, Coughlan, and Savaskan (2011)	✓	✓							✓				✓		✓	✓			✓					
Su (2009)	✓	✓	✓					✓	✓						✓	✓	✓					✓	✓	✓
Swinney (2011)	✓		✓						✓														✓	
Ülkü et al. (2013)	✓		✓	✓						✓					✓									✓
Ülkü and Gürler (2018)	✓	✓		✓						✓					✓	✓								✓
Welling (1989)	✓						✓		✓						✓					✓				
Xiao et al. (2010)			✓			✓			✓						✓	✓	✓						✓	✓
Xu et al. (2015)	✓				✓				✓	✓					✓	✓	✓							✓
Xu et al. (2018)	✓								✓						✓	✓	✓							✓

(Continues)

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TABLE A2 (Continued)

Article	CB							RP							PE							RM		
	PV	CH	TC	TP	RT	DR	RA	G	M	T	F	S	X	O	PR	I	SC	FD	C	PC	PA	A	P	D
Yalabik et al. (2005)	✓		✓	✓					✓						✓								✓	✓
Yang et al. (2017)	✓		✓					✓				✓			✓						✓			✓
Yoo (2014)	✓					✓			✓											✓				✓
Yoo et al. (2015)						✓			✓						✓		✓							✓
Zhang (2013)	✓		✓						✓						✓									✓

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TABLE A3 Classification of the empirical literature

Article	CB				RP							PE							RM		
	CR	AR	BI	BA	G	M	T	F	S	X	O	PR	I	SC	FD	C	PC	PA	A	P	D
Anderson et al. (2009a)				✓	✓							✓					✓				
Autry (2005)											✓									✓	
Bahn and Boyd (2014)		✓					✓											✓			
Bonifield et al. (2010)	✓		✓	✓							✓						✓				
Bower and Maxham III (2012)	✓	✓		✓		✓															
d'Astous and Guèvremont (2008)	✓						✓			✓											
Davis et al. (1998)											✓	✓						✓			✓
Gelbrich et al. (2017)			✓			✓															
Hawes and Lumpkin (1986)	✓				✓																
Heim and Field (2007)	✓					✓	✓	✓		✓											
Heiman et al. (2001)	✓				✓												✓				
Heiman et al. (2015)			✓				✓			✓											
Hjort and Lantz (2016)				✓		✓															
Hsieh (2013)	✓		✓								✓										
Janakiraman and Ordóñez (2012)	✓			✓			✓	✓													
Janakiraman et al. (2016)			✓	✓		✓	✓	✓	✓	✓					✓		✓	✓			
Jeng (2017)	✓		✓								✓										
Kim and Wansink (2012)	✓								✓												
Lantz and Hjort (2013)				✓		✓															
Mixon (1999)					✓							✓									
Mollenkopf et al. (2007)	✓		✓					✓													
Oghazi et al. (2018)	✓		✓								✓										
Pei et al. (2014)	✓		✓			✓															
Petersen and Kumar (2010)				✓							✓										
Posselt et al. (2008)		✓				✓	✓					✓									
Powers and Jack (2013)	✓	✓		✓							✓										
Rao et al. (2018)				✓			✓														
Seo et al. (2016)			✓		✓																
Shang, Pekgün, et al. (2017)				✓	✓											✓					✓
Suwelack et al. (2011)	✓	✓	✓				✓	✓									✓				
Van den Poel and Leunis (1999)			✓		✓							✓									
Wang (2009)				✓	✓		✓														
Wood (2001)	✓			✓		✓			✓												
Zhang et al. (2017)	✓		✓			✓	✓														
Zhou and Hinz (2016)				✓							✓										

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